ANNEX I

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT

Tysabri 300 mg concentrate for solution for infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each mL of concentrate contains 20 mg of natalizumab.

When diluted (see section 6.6), the solution for infusion contains approximately 2.6 mg per mL of natalizumab.

Natalizumab is a recombinant humanised anti-α4-integrin antibody produced in a murine cell line by recombinant DNA technology.

Excipients with known effect

Each vial contains 2.3 mmol (or 52 mg) sodium and 3 mg polysorbate 80 (see section 4.4 for further information).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Concentrate for solution for infusion.

Colourless, clear to slightly opalescent solution with a pH of 5.8 – 6.4 and an osmolality of 268 – 308 mOsm/kg.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Tysabri is indicated as single disease modifying therapy in adults with highly active relapsing remitting multiple sclerosis (RRMS) for the following patient groups:

- Patients with highly active disease despite a full and adequate course of treatment with at least one disease modifying therapy (DMT) (for exceptions and information about washout periods see sections 4.4 and 5.1)

or

- Patients with rapidly evolving severe RRMS defined by 2 or more disabling relapses in one year, and with 1 or more Gadolinium enhancing lesions on brain Magnetic Resonance Imaging (MRI) or a significant increase in T2 lesion load as compared to a previous recent MRI.

4.2 Posology and method of administration

Therapy is to be initiated and continuously supervised by specialised physicians experienced in the diagnosis and treatment of neurological conditions, in centres with timely access to MRI.

Patients treated with this medicinal product must be given the Patient Card and be informed about the risks of the medicinal product (see also package leaflet). After 2 years of treatment, patients should be re-informed about the risks, especially the increased risk of Progressive Multifocal Leukoencephalopathy (PML), and should be instructed together with their caregivers on early signs and symptoms of PML.

Resources for the management of hypersensitivity reactions and access to MRI should be available.

Some patients may have been exposed to immunosuppressive medicinal products (e.g. mitoxantrone, cyclophosphamide, azathioprine). These medicinal products have the potential to cause prolonged immunosuppression, even after dosing is discontinued. Therefore, the physician must confirm that such patients are not immunocompromised before starting treatment (see section 4.4).

Posology

Tysabri 300 mg is administered by intravenous infusion once every 4 weeks.

Continued therapy must be carefully reconsidered in patients who show no evidence of therapeutic benefit beyond 6 months.

Data on the safety and efficacy of natalizumab at 2 years were generated from controlled, double–blind studies. After 2 years continued therapy should be considered only following a reassessment of the potential for benefit and risk. Patients should be re-informed about the risk factors for PML, like duration of treatment, immunosuppressant use prior to receiving the medicinal product and the presence of anti-John Cunningham virus (JCV) antibodies (see section 4.4).

Readministration

The efficacy of re-administration has not been established (for safety see section 4.4).

Special populations

Elderly

This medicinal product is not recommended for use in patients aged over 65 years due to a lack of data in this population.

Renal and hepatic impairment

Studies have not been conducted to examine the effects of renal or hepatic impairment.

The mechanism for elimination and results from population pharmacokinetics suggest that dose adjustment would not be necessary in patients with renal or hepatic impairment.

Paediatric population

The safety and efficacy of this medicinal product in children and adolescents up to 18 years have not been established. Currently available data are described in sections 4.8 and 5.1.

Method of administration

This medicinal product is for intravenous use.

For instructions on dilution of the medicinal product before administration (see section 6.6).

After dilution (see section 6.6), the infusion is to be administered over approximately 1 hour, and patients are to be observed during the infusion and for 1 hour after the completion of the infusion for signs and symptoms of hypersensitivity reactions.

After the first 12 intravenous Tysabri doses, patients should continue to be observed during infusion. If the patients have not experienced any infusion reactions, the post dose observation time may be reduced or removed according to clinical judgement.

Patients restarting natalizumab treatment after a treatment gap ≥ 6 months are to be observed during the infusion and for 1 hour after the completion of the infusion for signs and symptoms of hypersensitivity reactions for the first 12 intravenous infusions after restarting therapy.

Tysabri 300 mg concentrate for solution for infusion must not be administered as a bolus injection.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

Progressive multifocal leukoencephalopathy (PML).

Patients with increased risk for opportunistic infections, including immunocompromised patients (including those currently receiving immunosuppressive therapies or those immunocompromised by prior therapies (see sections 4.4 and 4.8)).

Combination with other DMTs.

Known active malignancies, except for patients with cutaneous basal cell carcinoma.

4.4 Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and batch number of the administered product should be clearly recorded.

Progressive Multifocal Leukoencephalopathy (PML)

Use of this medicinal product has been associated with an increased risk of PML, an opportunistic infection caused by JC virus, which may be fatal or result in severe disability. Due to this increased risk of developing PML, the benefits and risks of treatment should be individually reconsidered by the specialist physician and the patient; patients must be monitored at regular intervals throughout and should be instructed together with their caregivers on early signs and symptoms of PML. JC virus also causes JCV granule cell neuronopathy (GCN) which has been reported in patients treated with this medicinal product. Symptoms of JCV GCN are similar to symptoms of PML (i.e. cerebellar syndrome).

The following risk factors are associated with an increased risk of PML:

- The presence of anti-JCV antibodies.
- Treatment duration, especially beyond 2 years. After 2 years all patients should be re-informed about the risk of PML with the medicinal product.
- Immunosuppressant use prior to receiving the medicinal product.

Patients who are anti-JCV antibody positive are at an increased risk of developing PML compared to patients who are anti-JCV antibody negative. Patients who have all three risk factors for PML (i.e., are anti-JCV antibody positive **and** have received more than 2 years of therapy with this medicinal product **and** have received prior immunosuppressant therapy) have a significantly higher risk of PML.

In anti-JCV antibody positive natalizumab treated patients who have not used prior immunosuppressants the level of anti-JCV antibody response (index) is associated with the level of risk for PML.

In anti-JCV antibody positive patients, extended interval dosing of Tysabri (average dosing interval of approximately 6 weeks) is suggested to be associated with a lower PML risk compared to approved dosing. If utilising extended interval dosing, caution is required because the efficacy of extended interval dosing has not been established and the associated benefit/risk balance is currently unknown (see section 5.1, *Intravenous administration Q6W*). For further information, refer to the Healthcare Professional Guide.

Patients considered at high risk with this treatment should only continue the treatment if the benefits outweigh the risks. For the estimation of PML risk in the different patient subgroups, please refer to the Healthcare Professional Guide.

Anti-JCV antibody testing

Anti-JCV antibody testing provides supportive information for risk stratification of treatment with this medicinal product. Testing for serum anti-JCV antibody prior to initiating therapy or in patients receiving the medicinal product with an unknown antibody status is recommended. Anti-JCV antibody negative patients may still be at risk of PML for reasons such as a new JCV infection, fluctuating antibody status or a false negative test result. Re-testing of anti-JCV antibody negative patients every 6 months is recommended. Retesting low index patients who have no history of prior immunosuppressant use every 6 months once they reach the 2-year treatment point is recommended.

The anti-JCV antibody assay (ELISA) should not be used to diagnose PML. Use of plasmapheresis/plasma exchange (PLEX) or intravenous immunoglobulin (IVIg) can affect meaningful interpretation of serum anti-JCV antibody testing. Patients should not be tested for anti-JCV antibodies within 2 weeks of PLEX due to removal of antibodies from the serum, or within 6 months of IVIg (i.e. 6 months = 5 half-lives for immunoglobulins).

For further information on anti-JCV antibody testing please see the Healthcare Professional Guide.

MRI screening for PML

Before initiation of treatment with this medicinal product, a recent (usually within 3 months) MRI should be available as a reference and be repeated at least on a yearly basis. More frequent MRIs (e.g. on a 3 to 6 monthly basis) using an abbreviated protocol should be considered for patients at higher risk of PML. This includes:

- Patients who have all three risk factors for PML (i.e., are-anti-JCV antibody positive **and** have received more than 2 years of therapy with this medicinal product **and** have received prior immunosuppressant therapy),

or

- Patients with a high anti-JCV antibody index who have received more than 2 years of therapy with this medicinal product and without prior history of immunosuppressant therapy.

Current evidence suggests that the risk of PML is low at an index equal to or below 0.9 and increases substantially above 1.5 for patients who have been on treatment with this medicinal product for longer than 2 years (see the Healthcare Professional Guide for further information).

PML should be considered as a differential diagnosis in any MS patient taking Tysabri presenting with neurological symptoms and/or new brain lesions in MRI. Cases of asymptomatic PML based on MRI and positive JCV DNA in the cerebrospinal fluid have been reported.

Physicians should refer to the Healthcare Professional Guide for further information on managing the risk of PML in natalizumab-treated patients.

If PML or JCV GCN is suspected, further dosing must be suspended until PML has been excluded.

The specialised physician should evaluate the patient to determine if the symptoms are indicative of neurological dysfunction and, if so, whether these symptoms are typical of MS or possibly suggestive of PML or JCV GCN. If any doubt exists, further evaluation, including MRI scan preferably with contrast (compared with pre-treatment baseline MRI), CSF testing for JC Viral DNA and repeat neurological assessments, should be considered as described in the Healthcare Professional Guide (see Educational guidance). Once the clinician has excluded PML and/or JCV GCN (if necessary, by repeating clinical, imaging and/or laboratory investigations if clinical suspicion remains), dosing may resume.

The physician should be particularly alert to symptoms suggestive of PML or JCV GCN that the patient may not notice (e.g. cognitive, psychiatric symptoms or cerebellar syndrome). Patients should also be advised to inform their partner or caregivers about their treatment, since they may notice symptoms that the patient is not aware of.

PML has been reported following discontinuation of this medicinal product in patients who did not have findings suggestive of PML at the time of discontinuation. Patients and physicians should continue to follow the same monitoring protocol and be alert for any new signs or symptoms that may be suggestive of PML for approximately 6 months following discontinuation of Tysabri.

If a patient develops PML the dosing of natalizumab must be permanently discontinued.

Following reconstitution of the immune system in immunocompromised patients with PML improved outcome has been seen.

Based on a retrospective analysis of natalizumab-treated patients since its approval, no difference was observed on 2-year survival after PML diagnosis between patients who received PLEX and those who did not. For other considerations on the management of PML, see the Healthcare Professional Guide.

PML and IRIS (Immune Reconstitution Inflammatory Syndrome)

IRIS occurs in almost all PML patients treated with this medicinal product after withdrawal or removal of the medicinal product. IRIS is thought to result from the restoration of immune function in patients with PML, which can lead to serious neurological complications and may be fatal. Monitoring for development of IRIS and appropriate treatment of the associated inflammation during recovery from PML should be undertaken (see the Healthcare Professional Guide for further information).

Infections including other opportunistic infections

Other opportunistic infections have been reported with use of this medicinal product, primarily in patients with Crohn’s disease who were immunocompromised or where significant co-morbidity existed, however increased risk of other opportunistic infections with use of the medicinal product in patients without these co-morbidities cannot currently be excluded. Opportunistic infections were also detected in MS patients treated with this medicinal product as a monotherapy (see section 4.8).

This treatment increases the risk of developing encephalitis and meningitis caused by herpes simplex and varicella zoster viruses. Serious, life-threatening, and sometimes fatal cases have been reported in the post-marketing setting in multiple sclerosis patients receiving the treatment (see section 4.8). If herpes encephalitis or meningitis occurs, the medicinal product should be discontinued, and appropriate treatment for herpes encephalitis or meningitis should be administered.

Acute retinal necrosis (ARN) is a rare fulminant viral infection of the retina caused by the family of herpes viruses (e.g. varicella zoster). ARN has been observed in patients being administered this medicinal product and can be potentially blinding. Patients presenting with eye symptoms such as decreased visual acuity, redness and painful eye should be referred for retinal screening for ARN. Following clinical diagnosis of ARN, discontinuation of this medicinal product should be considered in these patients.

Prescribers should be aware of the possibility that other opportunistic infections may occur during therapy and should include them in the differential diagnosis of infections that occur in natalizumab-treated patients. If an opportunistic infection is suspected, dosing is to be suspended until such infections can be excluded through further evaluations.

If a patient receiving this medicinal product develops an opportunistic infection, dosing of the medicinal product must be permanently discontinued.

Educational guidance

All physicians who intend to prescribe the medicinal product must ensure they are familiar with the Healthcare Professional Guide.

Physicians must discuss the benefits and risks of natalizumab therapy with the patient and provide them with a Patient Card. Patients should be instructed that if they develop any infection then they should inform their physician that they are being treated with this medicinal product.

Physicians should counsel patients on the importance of uninterrupted dosing, particularly in the early months of treatment (see Hypersensitivity).

Hypersensitivity

Hypersensitivity reactions have been associated with this medicinal product, including serious systemic reactions (see section 4.8). These reactions usually occurred during the infusion or up to 1 hour after completion of the infusion. The risk for hypersensitivity was greatest with early infusions and in patients re-exposed to treatment following an initial short exposure (one or two infusions) and extended period (three months or more) without treatment. However, the risk of hypersensitivity reactions should be considered for every infusion administered.

Patients are to be observed during the infusion and for 1 hour after the completion of the infusion (see section 4.8). Resources for the management of hypersensitivity reactions should be available.

This product should be discontinued and appropriate therapy initiated at the first symptoms or signs of hypersensitivity.

Patients who have experienced a hypersensitivity reaction must be permanently discontinued from treatment with natalizumab.

Concurrent treatment with immunosuppressants

The safety and efficacy of this medicinal product in combination with other immunosuppressive and antineoplastic therapies have not been fully established. Concurrent use of these agents with this medicinal product may increase the risk of infections, including opportunistic infections, and is contraindicated (see section 4.3).

In phase 3 MS clinical trials with natalizumab intravenous infusion, concomitant treatment of relapses with a short course of corticosteroids was not associated with an increased rate of infection. Short courses of corticosteroids can be used in combination with this medicinal product.

Prior treatment with immunosuppressive or immunomodulatory therapies

Patients with a treatment history of immunosuppressant medications are at increased risk for PML.

Data from an observational study demonstrated that there is no increased risk of PML for the group of patients switching to natalizumab from fingolimod, dimethyl fumarate, or teriflunomide when compared to the group of patients switching from either beta interferon or glatiramer acetate.

No studies have been performed to evaluate the safety of natalizumab when switching patients from DMTs other than beta interferon, glatiramer acetate, fingolimod, dimethyl fumarate and teriflunomide. It is unknown if patients switching from other therapies to natalizumab have an increased risk of PML compared to those switching from the above-mentioned DMTs, therefore these patients should be monitored more frequently (i.e. similarly to patients switching from immunosuppressants to natalizumab).

Care should be taken with patients who have previously received immunosuppressants to allow sufficient time for immune function recovery to occur. Physicians must evaluate each individual case to determine whether there is evidence of an immunocompromised state prior to commencing treatment (see section 4.3).

When switching patients from another DMT to this medicinal product, the half-life and mode of action of the other therapy must be considered to avoid an additive immune effect whilst at the same time minimising the risk of disease reactivation. A Complete Blood Count (CBC, including lymphocytes) is recommended prior to initiating treatment to ensure that immune effects of the previous therapy (i.e. cytopenia) have resolved.

Patients can switch directly from beta interferon or glatiramer acetate to natalizumab providing there are no signs of relevant treatment-related abnormalities e.g. neutropenia and lymphopenia.

When switching from dimethyl fumarate, the washout period should be sufficient for lymphocyte count to recover before treatment is started.

Following discontinuation of fingolimod, lymphocyte count progressively returns to normal range within 1 to 2 months after stopping therapy. The washout period should be sufficient for lymphocyte count to recover before treatment is started.

Teriflunomide is eliminated slowly from the plasma. Without an accelerated elimination procedure, clearance of teriflunomide from plasma can take from several months up to 2 years. An accelerated elimination procedure as defined in the teriflunomide Summary of Product Characteristics is recommended or alternatively washout period should not be shorter than 3.5 months. Caution regarding potential concomitant immune effects is required when switching patients from teriflunomide to this medicinal product.

Alemtuzumab has profound prolonged immunosuppressive effects. As the actual duration of these effects is unknown, initiating treatment with this medicinal product after alemtuzumab is not recommended unless the benefits clearly outweigh the risks for the individual patient.

Immunogenicity

Disease exacerbations or infusion related events may indicate the development of antibodies against natalizumab. In these cases, the presence of antibodies should be evaluated and if these remain positive in a confirmatory test after at least 6 weeks, treatment should be discontinued, as persistent antibodies are associated with a substantial decrease in efficacy of this medicinal product and an increased incidence of hypersensitivity reactions (see section 4.8).

Since patients who have received an initial short exposure to this medicinal product and then had an extended period without treatment are at a higher risk of developing anti-natalizumab antibodies and/or hypersensitivity upon redosing, the presence of antibodies should be evaluated and if these remain positive in a confirmatory test after at least 6 weeks, the patient should not receive further treatment with natalizumab (see section 5.1).

Hepatic events

Spontaneous serious adverse reactions of liver injury have been reported during the post-marketing phase (see section 4.8). These liver injuries may occur at any time during treatment, even after the first

dose. In some instances, the reaction reoccurred when treatment was reintroduced. Some patients with a past medical history of an abnormal liver test have experienced an exacerbation of abnormal liver test while on treatment. Patients should be monitored as appropriate for impaired liver function and be instructed to contact their physician in case signs and symptoms suggestive of liver injury occur, such as jaundice and vomiting. In cases of significant liver injury this medicinal product should be discontinued.

Thrombocytopenia

Thrombocytopenia, including immune thrombocytopenic purpura (ITP), has been reported with the use of natalizumab. Delay in the diagnosis and treatment of thrombocytopenia may lead to serious and life-threatening sequelae. Patients should be instructed to report to their physician immediately if they experience any signs of unusual or prolonged bleeding, petechiae, or spontaneous bruising. If thrombocytopenia is identified, discontinuation of natalizumab should be considered.

Stopping therapy

If a decision is made to stop treatment with natalizumab, the physician needs to be aware that natalizumab remains in the blood and has pharmacodynamic effects (e.g increased lymphocyte counts) for approximately 12 weeks following the last dose. Starting other therapies during this interval will result in a concomitant exposure to natalizumab. For medicinal products such as interferon and glatiramer acetate, concomitant exposure of this duration was not associated with safety risks in clinical trials. No data are available in MS patients regarding concomitant exposure with immunosuppressant medication. Use of these medicinal products soon after the discontinuation of natalizumab may lead to an additive immunosuppressive effect. This should be carefully considered on a case-by-case basis, and a wash-out period of natalizumab might be appropriate. Short courses of steroids used to treat relapses were not associated with increased infections in clinical trials.

Polysorbate 80 (E 433) content

This medicinal product contains 3 mg of polysorbate 80 in each vial. Polysorbates may cause allergic reactions.

Sodium content

Before dilution, this medicinal product contains 52 mg sodium per vial of medicinal product, equivalent to 2.6% of the WHO recommended maximum daily intake of 2 g sodium for an adult. When diluted in 100 ml sodium chloride 9 mg/ml (0.9%), this medicinal product contains 406 mg sodium per dose. To be taken into consideration in case patients are on a controlled sodium diet.

4.5 Interaction with other medicinal products and other forms of interaction

Natalizumab is contraindicated in combination with other DMTs (see section 4.3).

Immunisations

In a randomised, open label study of 60 patients with relapsing MS there was no significant difference in the humoral immune response to a recall antigen (tetanus toxoid) and only slightly slower and reduced humoral immune response to a neoantigen (keyhole limpet haemocyanin) was observed in patients who were treated with this medicinal product for 6 months compared to an untreated control group. Live vaccines have not been studied.

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

If a woman becomes pregnant while taking this medicinal product, discontinuation should be considered. A benefit/risk evaluation of the use of this medicinal product during pregnancy should take into account the patient's clinical condition and the possible return of disease activity after stopping the medicinal product.

Pregnancy

Studies in animals have shown reproductive toxicity (see section 5.3).

Data from clinical trials, a prospective pregnancy registry, post-marketing cases and available literature do not suggest an effect of natalizumab exposure on pregnancy outcomes.

The completed prospective Tysabri pregnancy registry contained 355 pregnancies with available outcomes. There were 316 live births, 29 of which were reported to have birth defects. Sixteen of the 29 were classified as major defects. The rate of defects corresponds to the defect rates reported in other pregnancy registries involving MS patients. There is no evidence of a specific pattern of birth defects with this medicinal product.

There are no adequate and well-controlled studies of natalizumab therapy in pregnant women.

Thrombocytopenia and anaemia in infants born to women exposed to natalizumab during pregnancy were reported in the post-marketing setting. Monitoring of platelet counts, haemoglobin, and haematocrit is recommended in neonates born to women exposed to natalizumab during pregnancy.

This medicinal product should be used during pregnancy only if clearly needed. If a woman becomes pregnant while taking natalizumab, discontinuation of natalizumab should be considered.

Breast-feeding

Natalizumab is excreted in human milk. The effect of natalizumab on newborn/infants is unknown. Breast-feeding should be discontinued during treatment with natalizumab.

Fertility

Reductions in female guinea pig fertility were observed in one study at doses in excess of the human dose; natalizumab did not affect male fertility. It is considered unlikely that natalizumab will affect fertility performance in humans following the maximum recommended dose.

4.7 Effects on ability to drive and use machines

Tysabri has a minor influence on the ability to drive and use machines. Dizziness may occur following administration of this medicinal product (see section 4.8).

4.8 Undesirable effects

Summary of the safety profile

In placebo-controlled trials in 1 617 MS patients treated with natalizumab for up to 2 years (placebo: 1 135), adverse events leading to discontinuation of therapy occurred in 5.8% of patients treated with natalizumab (placebo: 4.8%). Over the 2-year duration of the studies, 43.5% of patients treated with natalizumab reported adverse reactions (placebo: 39.6%).

In clinical trials in 6 786 patients treated with natalizumab (intravenous infusion and subcutaneous injection), the most frequently occurring adverse reactions were headache (32%), nasopharyngitis

(27%), fatigue (23%), urinary tract infection (16%), nausea (15%), arthralgia (14%), and dizziness (11%) associated with natalizumab administration.

Tabulated list of adverse reactions

Adverse reactions arising from clinical studies, post-authorisation safety studies and spontaneous reports are presented in Table 1, below. Within the system organ classes they are listed under the following headings: Very common (≥1/10); Common (≥1/100 to <1/10); Uncommon (≥1/1 000 to <1/100); Rare (≥1/10 000 to <1/1 000); Very rare (<1/10 000), not known (cannot be estimated from the available data). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1. Adverse reactions

MedDRA System Organ Class	Frequency of adverse reactions				
	Very Common	*Common*	*Uncommon*	*Rare*	*Not known*
Infections and infestations	Nasopharyngitis Urinary tract infection	Herpes infection	Progressive multifocal leukoencephalopathy	Herpes ophthalmic	Meningoencephalitis herpetic JC virus granule cell neuropathy Necrotising herpetic retinopathy
Blood and lymphatic system disorders		Anaemia	Thrombocytopenia, Immune thrombocytopenic purpura (ITP) Eosinophilia	Haemolytic anaemia Nucleated red cells	
Immune system disorders		Hypersensitivity	Anaphylactic reaction Immune reconstitution inflammatory syndrome		
Nervous system disorders	Dizziness Headache				
Vascular disorders		Flushing			
Respiratory, thoracic and mediastinal disorders		Dyspnoea			
Gastrointestinal disorders	Nausea	Vomiting			
Hepatobiliary disorders				Hyperbilirubinaemia	Liver injury
Skin and subcutaneous tissue disorders		Pruritus Rash Urticaria		Angioedema	
Musculoskeletal and connective tissue disorders	Arthralgia				
General disorders and administration site conditions	Fatigue	Pyrexia Chills Infusion site reaction Injection site reaction	Face oedema		

MedDRA System Organ Class	Frequency of adverse reactions				
	Very Common	*Common*	*Uncommon*	*Rare*	*Not known*
Investigations		Hepatic enzyme increased Drug specific antibody present			
Injury, poisoning and procedural complications	Infusion related reaction				

Description of selected adverse reactions

Infusion-related reactions (IRR)

In 2-year controlled clinical trials in MS patients, an infusion-related event was defined as an adverse event occurring during the infusion or within 1 hour of the completion of the infusion. These occurred in 23.1% of MS patients treated with natalizumab (placebo: 18.7%). Events reported more commonly with natalizumab than with placebo included dizziness, nausea, urticaria and rigors.

Hypersensitivity reactions

In 2-year controlled clinical trials in MS patients, hypersensitivity reactions occurred in up to 4% of patients. Anaphylactic/anaphylactoid reactions occurred in less than 1% of patients receiving this medicinal product. Hypersensitivity reactions usually occurred during the infusion or within the 1-hour period after the completion of the infusion (see section 4.4). In post-marketing experience, there have been reports of hypersensitivity reactions which have occurred with one or more of the following associated symptoms: hypotension, hypertension, chest pain, chest discomfort, dyspnoea, angioedema, in addition to more usual symptoms such as rash and urticaria.

Immunogenicity

In 10% of patients, antibodies against natalizumab were detected in 2-year controlled clinical trials in MS patients. Persistent anti-natalizumab antibodies (one positive test reproducible on retesting at least 6 weeks later) developed in approximately 6% of patients. Antibodies were detected on only one occasion in an additional 4% of patients. Persistent antibodies were associated with a substantial decrease in the effectiveness of natalizumab and an increased incidence of hypersensitivity reactions. Additional infusion-related reactions associated with persistent antibodies included rigors, nausea, vomiting and flushing (see section 4.4).

If, after approximately 6 months of therapy, persistent antibodies are suspected, either due to reduced efficacy or due to occurrence of infusion-related events, they may be detected and confirmed with a subsequent test 6 weeks after the first positive test. Given that efficacy may be reduced, or the incidence of hypersensitivity or infusion-related reactions may be increased in a patient with persistent antibodies, treatment should be discontinued in patients who develop persistent antibodies.

Infections, including PML and opportunistic infections

In 2-year controlled clinical trials in MS patients, the rate of infection was approximately 1.5 per patient-year in both natalizumab- and placebo-treated patients. The nature of the infections was generally similar in natalizumab- and placebo-treated patients. A case of *cryptosporidium* diarrhoea was reported in MS clinical trials. In other clinical trials, cases of additional opportunistic infections have been reported, some of which were fatal. The majority of patients did not interrupt natalizumab therapy during infections and recovery occurred with appropriate treatment.

In clinical trials, herpes infections (Varicella-Zoster virus, Herpes-simplex virus) occurred slightly more frequently in natalizumab-treated patients than in placebo-treated patients. In post-marketing experience, serious, life-threatening, and sometimes fatal cases of encephalitis and meningitis caused by herpes simplex or varicella zoster have been reported in multiple sclerosis patients receiving natalizumab. The duration of treatment with natalizumab prior to onset ranged from a few months to several years (see section 4.4).

In post-marketing experience, rare cases of ARN have been observed in patients receiving this medicinal product. Some cases have occurred in patients with central nervous system (CNS) herpes infections (e.g. herpes meningitis and encephalitis). Serious cases of ARN, either affecting one or both eyes, led to blindness in some patients. The treatment reported in these cases included anti-viral therapy and, in some cases, surgery (see section 4.4).

Cases of PML have been reported from clinical trials, post-marketing observational studies and post-marketing passive surveillance. PML usually leads to severe disability or death (see section 4.4). Cases of JCV GCN have also been reported during post-marketing use of Tysabri. Symptoms of JCV GCN are similar to PML.

Hepatic events

Spontaneous cases of serious liver injuries, increased liver enzymes, hyperbilirubinaemia have been reported during the post-marketing phase (see section 4.4).

Anaemia and haemolytic anaemia

Rare, serious cases of anaemia and haemolytic anaemia have been reported in patients treated with this medicinal product in post-marketing observational studies.

Effects on laboratory tests

In 2-year controlled clinical trials in MS patients, treatment with natalizumab was associated with increases in circulating lymphocytes, monocytes, eosinophils, basophils and nucleated red blood cells. Elevations in neutrophils were not seen. Increases from baseline for lymphocytes, monocytes, eosinophils and basophils ranged from 35% to 140% for individual cell types but mean cell counts remained within normal ranges with IV administration. During treatment with IV form of this medicinal product, small reductions in haemoglobin (mean decrease 0.6 g/dL), haematocrit (mean decrease 2%) and red blood cell counts (mean decrease 0.1 x 10^6/L) were seen. All changes in haematological variables returned to pre-treatment values, usually within 16 weeks of last dose of the medicinal product and the changes were not associated with clinical symptoms. In post-marketing experience, there have also been reports of eosinophilia (eosinophil count > 1 500/mm^3) without clinical symptoms. In such cases where therapy was discontinued the elevated eosinophil levels resolved.

Thrombocytopenia
In post-marketing experience, thrombocytopenia and immune thrombocytopenic purpura (ITP) have been reported with uncommon frequency.

Paediatric population

Serious adverse events were evaluated in 621 MS paediatric patients included in a meta-analysis (see also section 5.1). Within the limitations of these data, there were no new safety signals identified in this patient population. 1 case of herpes meningitis was reported in the meta-analysis. No cases of PML were identified in the meta-analysis, however, PML has been reported in natalizumab-treated paediatric patients in the post-marketing setting.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system listed in Appendix V.

4.9 Overdose

Safety of doses higher than 300 mg has not been adequately evaluated. The maximum amount of natalizumab that can be safely administered has not been determined.

There is no known antidote for natalizumab overdose. Treatment consists of discontinuation of the medicinal product and supportive therapy as needed.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Immunosuppressants, monoclonal antibodies, ATC code: L04AG03

Pharmacodynamic effects

Natalizumab is a selective adhesion-molecule inhibitor and binds to the α4-subunit of human integrins, which is highly expressed on the surface of all leukocytes, with the exception of neutrophils. Specifically, natalizumab binds to the α4β1 integrin, blocking the interaction with its cognate receptor, vascular cell adhesion molecule-1 (VCAM-1), and ligands osteopontin, and an alternatively spliced domain of fibronectin, connecting segment-1 (CS-1). Natalizumab blocks the interaction of α4β7 integrin with the mucosal addressin cell adhesion molecule-1 (MadCAM-1). Disruption of these molecular interactions prevents transmigration of mononuclear leukocytes across the endothelium into inflamed parenchymal tissue. A further mechanism of action of natalizumab may be to suppress ongoing inflammatory reactions in diseased tissues by inhibiting the interaction of α4-expressing leukocytes with their ligands in the extracellular matrix and on parenchymal cells. As such, natalizumab may act to suppress inflammatory activity present at the disease site, and inhibit further recruitment of immune cells into inflamed tissues.

In MS, lesions are believed to occur when activated T-lymphocytes cross the blood-brain barrier (BBB). Leukocyte migration across the BBB involves interaction between adhesion molecules on inflammatory cells and endothelial cells of the vessel wall. The interaction between α4β1 and its targets is an important component of pathological inflammation in the brain and disruption of these interactions leads to reduced inflammation. Under normal conditions, VCAM-1 is not expressed in the brain parenchyma. However, in the presence of pro-inflammatory cytokines, VCAM-1 is upregulated on endothelial cells and possibly on glial cells near the sites of inflammation. In the setting of central nervous system (CNS) inflammation in MS, it is the interaction of α4β1 with VCAM-1, CS-1 and osteopontin that mediates the firm adhesion and transmigration of leukocytes into the brain parenchyma and may perpetuate the inflammatory cascade in CNS tissue. Blockade of the molecular interactions of α4β1 with its targets reduces inflammatory activity present in the brain in MS and inhibits further recruitment of immune cells into inflamed tissue, thus reducing the formation or enlargement of MS lesions.

Clinical efficacy

AFFIRM clinical study

Efficacy as monotherapy has been evaluated in one randomised, double-blind, placebo-controlled study lasting 2 years (AFFIRM study) in RRMS patients who had experienced at least 1 clinical relapse during the year prior to entry and had a Kurtzke Expanded Disability Status Scale (EDSS) score between 0 and 5. Median age was 37 years, with a median disease duration of 5 years. The

patients were randomised with a 2:1 ratio to receive Tysabri 300 mg (n = 627) or placebo (n = 315) every 4 weeks for up to 30 infusions. Neurological evaluations were performed every 12 weeks and at times of suspected relapse. MRI evaluations for T1-weighted gadolinium (Gd)-enhancing lesions and T2-hyperintense lesions were performed annually.

Study features and results are presented in the Table 2.

Table 2. AFFIRM study: Main features and results		
Design	Monotherapy; randomised double-blind placebo-controlled parallel-group trial for 120 weeks	
Subjects	RRMS (McDonald criteria)	
Treatment	Placebo / Natalizumab 300 mg i.v. every 4 weeks	
One year endpoint	Relapse rate	
Two year endpoint	Progression on EDSS	
Secondary endpoints	Relapse rate derived variables / MRI-derived variables	
Subjects	Placebo	Natalizumab
Randomised	315	627
Completing 1 years	296	609
Completing 2 years	285	589
Age yrs, median (range)	37 (19-50)	36 (18-50)
MS-history yrs, median (range)	6.0 (0-33)	5.0 (0-34)
Time since diagnosis, yrs median (range)	2.0 (0-23)	2.0 (0-24)
Relapses in previous 12 months, median (range)	1.0 (0-5)	1.0 (0-12)
EDSS-baseline, median (range)	2 (0-6.0)	2 (0-6.0)
RESULTS		
Annual relapse rate		
After one year (primary endpoint)	0.805	0.261
After two years	0.733	0.235
One year	Rate ratio 0.33 $CI_{95\%}$ 0.26 ; 0.41	
Two years	Rate ratio 0.32 $CI_{95\%}$ 0.26 ; 0.40	
Relapse free		
After one year	53%	76%
After two years	41%	67%
Disability		
Proportion progressed[1](12-week confirmation; primary outcome)	29%	17%
	Hazard ratio 0.58, $CI_{95\%}$ 0.43; 0.73, p<0.001	
Proportion progressed[1](24-week confirmation)	23%	11%
	Hazard ratio 0.46, $CI_{95\%}$ 0.33; 0.64, p<0.001	
MRI (0-2 years)		
Median % change in T2-hyperintense lesion volume	+8.8%	-9.4% (p<0.001)
Mean number of new or newly-enlarging T2-hyperintense lesions	11.0	1.9 (p<0.001)
Mean number of T1-hypointense lesions	4.6	1.1 (p<0.001)
Mean number of Gd-enhancing lesions	1.2	0.1 (p<0.001)

Table 2. AFFIRM study: Main features and results
[1] Progression of disability was defined as at least a 1.0 point increase on the EDSS from a baseline EDSS >= 1.0 sustained for 12 or 24 weeks or at least a 1.5 point increase on the EDSS from a baseline EDSS = 0 sustained for 12 or 24 weeks.

In the sub-group of patients indicated for treatment of rapidly evolving RRMS (patients with 2 or more relapses and 1 or more Gd+ lesion), the annualised relapse rate was 0.282 in the natalizumab-treated group (n = 148) and 1.455 in the placebo group (n = 61) (p <0.001). Hazard ratio for disability progression was 0.36 (95% CI: 0.17, 0.76) p = 0.008. These results were obtained from a *post hoc* analysis and should be interpreted cautiously. No information on the severity of the relapses before inclusion of patients in the study is available.

Tysabri Observational Program (TOP, IMA-06-02)

The Tysabri Observational Program (TOP, IMA-06-02) was a multicentre, single-arm study initiated in 2007 to assess real-world safety and effectiveness outcomes in MS patients treated with Tysabri, with patient follow-up data for up to approximately 15 years. The study generated data from 6 319 patients in 17 countries, including 1 145 patients with up to 10 years of exposure and 102 patients with up to 15 years of exposure.

Overall, the safety findings of the TOP analyses were consistent with the known safety profile of natalizumab. Patients showed reductions in pre-treatment annualised relapse rate (ARR) regardless of the number of prior relapses, baseline EDSS, prior immunosuppressants use, or the number of DMTs used prior to natalizumab initiation. In the overall population, the ARR was 0.17 (95% CI: 0.17, 0.18) over the course of 15 years of follow up. Mean EDSS scores were similar from baseline (3.5, SD = 1.61) to year 15 (3.4, SD = 1.97) in patients treated with natalizumab.

A total of 5 635 patients had received another DMT prior to natalizumab initiation. The patients switching from beta interferon, glatiramer acetate or fingolimod had similar effectiveness outcomes to the overall natalizumab-treated population.

Paediatric population

A post-marketing meta-analysis was conducted using data from 621 paediatric MS patients treated with natalizumab (median age 17 years, range was 7 to 18 years, 91% aged ≥14 years). Within this analysis, a limited subset of patients with data available prior to treatment (158 of the 621 patients) demonstrated a reduction in ARR from 1.466 (95% CI 1.337, 1.604) prior to treatment to 0.110 (95% CI 0.094, 0.128).

Extended interval dosing

In a pre-specified, retrospective analysis of US anti-JCV antibody positive Tysabri patients intravenously administered (TOUCH Prescribing Program), the risk of PML was compared between patients treated with the approved dosing interval and patients treated with extended interval dosing as identified in the last 18 months of exposure (EID, average dosing intervals of approximately 6 weeks). The majority (85%) of patients dosed with EID had received the approved dosing for ≥ 1 year prior to switching to EID. The analysis showed a lower risk of PML in patients treated with EID (hazard ratio = 0.06, 95% CI of hazard ratio = 0.01 to 0.22).

Efficacy has been modelled for patients who switch to longer dosing after ≥1 year of approved dosing with this medicinal product under intravenous administration and who did not experience a relapse in the year prior to switching. Current pharmacokinetic/pharmacodynamic statistical modelling and simulation indicate that the risk of MS disease activity for patients switching to longer dosing intervals may be higher for patients with dosing intervals ≥ 7 weeks. No prospective clinical studies have been completed to validate these findings.

The efficacy of natalizumab when administered with EID has not been established; therefore, the benefit/risk balance of EID is unknown (see "*Intravenous administration Q6W"*).

Intravenous administration Q6W

Efficacy and safety were evaluated in a prospective, randomized, interventional, controlled, open-label, rater-blinded, international phase 3 study (NOVA, 101MS329), involving subjects with relapsing-remitting MS according to the 2017 McDonald criteria dosed intravenously every 6 weeks with natalizumab. The study was designed to estimate an efficacy difference between Q6W and Q4W dosing regimens.

The study randomized 499 subjects aged 18-60, with an EDSS score ≤ 5.5 at screening, who received at least 1 year of natalizumab treatment IV Q4W and were clinically stable (no relapse in the last 12 months, no gadolinium (Gd) enhancing T1 lesions at screening). In the study, subjects who switched to Q6W after at least one year of IV Q4W treatment with natalizumab were evaluated in relation to subjects who continued on IV Q4W treatment.

Baseline demographic subgroups of age, sex, duration of natalizumab exposure, country, body weight, anti-JCV status and number of relapses in the year prior to the first dose, number of relapses while on natalizumab, number of prior DMTs, and type of prior DMT were similar between the Q6W and Q4W dosing treatment arms.

Table 3. NOVA study: Main features and results		
Design	Monotherapy; phase 3b prospective, randomized, interventional, controlled, open-label, rater-blinded, international study	
Subjects	RRMS (McDonald criteria)	
Treatment administration (part 1)	Natalizumab Q4W 300 mg I.V.	Natalizumab Q6W 300 mg I.V.
Randomized	248	251
RESULTS		
mITT[a] population for part 1 at week 72	242	247
New/newly enlarging (N/NE) T2 lesions from baseline to Week 72 Subjects with number of lesions = 0	189 (78.1%)	202 (81.8%)
= 1	7 (3.6%)	5 (2.0%)
= 2	1 (0.5%)	2 (0.8%)
= 3	0	0
= 4	0	0
≥ 5	0	2* (0.8%)
missing	45 (18.6%)	36 (14.6%)
Adjusted mean N/NE T2-hyperintense lesions (primary endpoint)* 95% CI[b,c]	0.05 (0.01, 0.22)	0.20 (0.07, 0.63)
	p = 0.0755	
Proportion of subjects that developed N/NE T2 lesions	4.1%	4.3%
Proportion of subjects who developed T1-hypointense lesions	0.8%	1.2%
Proportion of subjects who developed Gd-enhancing lesions	0.4%	0.4%
Adjusted annualized relapse rate	0.00010	0.00013

Proportion of subjects free of relapse**	97.6%	96.9%
Proportion free of 24-week confirmed EDSS worsening	92%	90%

[a] mITT population, which included all randomized participants who received at least 1 dose of study treatment (natalizumab SID or natalizumab EID) and had at least 1 postbaseline result from the following clinical efficacy assessments: MRI efficacy assessments, relapses, EDSS, 9-HPT, T25FW, SDMT, TSQM, CGI scale.

[b] Estimated using negative binomial regression with treatment as classification and baseline body weight (≤ 80 vs > 80 kg), duration of natalizumab exposure at baseline (≤ 3 vs > 3 years), and region (North America, the UK, Europe and Israel, and Australia) as covariates.

[c] Observed lesions are included for analysis regardless of intercurrent events, and missing values due to efficacy or safety (6 subjects switched to Q4W dosing and 1 subject each on Q6W and Q4W dosing discontinued treatment) are imputed by the worst case of subjects on treatment at the same visit in the same treatment group or otherwise via multiple imputation.

* The numerical difference seen in the N/NE lesions between the two treatment groups was driven by a high number of lesions occurring in two subjects in the Q6W arm – one subject who developed lesions three months after treatment discontinuation and a second subject who was diagnosed with asymptomatic PML at week 72.

** Relapses – clinical relapses were assessed as defined by new or recurrent neurologic symptoms not associated with fever or infection having a minimum duration of 24 hours.

5.2 Pharmacokinetic properties

Following the repeat intravenous administration of a 300 mg dose of natalizumab to MS patients, the mean maximum observed serum concentration was 110 ± 52 µg/mL. Mean average steady-state trough natalizumab concentrations over the dosing period ranged from 23 µg/mL to 29 µg/mL in Q4W dosing. At any time, mean trough concentrations for the Q6W regimen were approximately 60 to 70% lower than for the Q4W regimen. The predicted time to steady state was approximately 24 weeks. Population pharmacokinetic analysis includes 12 studies and 1 781 subjects receiving doses ranging from 1 to 6 mg/kg and fixed doses of 150/300 mg.

Distribution

Median steady-state volume of distribution was 5.96 L (5.59-6.38 L, 95% confidence interval).

Elimination

Population median estimate for linear clearance was 6.08 mL/h (5.75-6.33 mL/h, 95% confidence interval) and the estimated median half-life was 28.2 days. The 95th percentile interval of the terminal half-life is from 11.6 to 46.2 days.

The population analysis of 1 781 patients explored the effects of selected covariates including body weight, age, gender, presence of anti-natalizumab antibodies and formulation on pharmacokinetics. Only body weight, the presence of anti-natalizumab antibodies and the formulation used in phase 2 studies were found to influence natalizumab disposition. Natalizumab clearance increased with body weight in a less-than-proportional manner, such that a +/-43% change in body weight resulted in only a -33% to 30% change in clearance. The presence of persistent anti-natalizumab antibodies increased natalizumab clearance approximately 2.45-fold, consistent with reduced serum natalizumab concentrations observed in persistently antibody-positive patients.

Special Populations

Paediatric population

The pharmacokinetics of natalizumab in paediatric MS patients has not been established.

Renal impairment

The pharmacokinetics of natalizumab in patients with renal insufficiency has not been studied.

Hepatic impairment

The pharmacokinetics of natalizumab in patients with hepatic insufficiency has not been studied.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity and genotoxicity.

Consistent with the pharmacological activity of natalizumab, altered trafficking of lymphocytes was seen as white blood cell increases as well as increased spleen weights in most *in vivo* studies. These changes were reversible and did not appear to have any adverse toxicological consequences.

In studies conducted in mice, growth and metastasis of melanoma and lymphoblastic leukaemia tumour cells was not increased by the administration of natalizumab.

No clastogenic or mutagenic effects of natalizumab were observed in the Ames or human chromosomal aberration assays. Natalizumab showed no effects on *in vitro* assays of α4-integrin-positive tumour line proliferation or cytotoxicity.

Reductions in female guinea pig fertility were observed in one study at doses in excess of the human dose; natalizumab did not affect male fertility.

The effect of natalizumab on reproduction was evaluated in 5 studies, 3 in guinea pigs and 2 in *cynomolgus* monkeys. These studies showed no evidence of teratogenic effects or effects on growth of offspring. In one study in guinea pigs, a small reduction in pup survival was noted. In a study in monkeys, the number of abortions was doubled in the natalizumab 30 mg/kg treatment groups versus matching control groups. This was the result of a high incidence of abortions in treated groups in the first cohort that was not observed in the second cohort. No effects on abortion rates were noted in any other study. A study in pregnant *cynomolgus* monkeys demonstrated natalizumab-related changes in the foetus that included mild anaemia, reduced platelet counts, increased spleen weights and reduced liver and thymus weights. These changes were associated with increased splenic extramedullary haematopoiesis, thymic atrophy and decreased hepatic haematopoiesis. Platelet counts were also reduced in offspring born to mothers treated with natalizumab until parturition, however there was no evidence of anaemia in these offspring. All changes were observed at doses in excess of the human dose and were reversed upon clearance of natalizumab.

In *cynomolgus* monkeys treated with natalizumab until parturition, low levels of natalizumab were detected in the breast milk of some animals.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sodium phosphate, monobasic, monohydrate
Sodium phosphate, dibasic, heptahydrate
Sodium chloride
Polysorbate 80 (E 433)
Water for injections

6.2 Incompatibilities

Tysabri 300 mg concentrate for solution for infusion must not be mixed with other medicinal products except those mentioned in section 6.6.

6.3 Shelf life

Unopened vial

4 years

Diluted solution

Chemical and physical stability of the diluted solution has been demonstrated for 72 hours at 2 °C – 8 °C and up to 30 °C.
From a microbiological point of view, after dilution with sodium chloride 9 mg/mL (0.9%) solution for injection, immediate use is recommended. If not used immediately, the diluted solution must be stored at 2 °C to 8 °C and infused within 24 hours of dilution. In-use storage times and conditions prior to use are the responsibility of the user.

6.4 Special precautions for storage

Store in a refrigerator (2 °C – 8 °C).
Do not freeze.
Keep the vial in the outer carton in order to protect from light.

For storage conditions after dilution of the medicinal product see section 6.3.

6.5 Nature and contents of container

15 mL concentrate in a vial (type I glass) with a stopper (chlorobutyl rubber) and a seal (aluminium) with a flip-off cap.

Pack size of one vial per carton.

6.6 Special precautions for disposal and other handling

Instructions for use:

- Inspect the vial for particles prior to dilution and administration. If particles are observed and/or the liquid in the vial is not colourless, clear to slightly opalescent, the vial must not be used.
- Use aseptic technique when preparing the solution for intravenous (IV) infusion. Remove flip-off cap from the vial. Insert the syringe needle into the vial through the centre of the rubber stopper and remove 15 mL concentrate for solution for infusion.
- Add the 15 mL concentrate for solution for infusion to 100 mL sodium chloride 9 mg/mL (0.9%) solution for injection. Gently invert the solution to mix completely. Do not shake.
- This medicinal product must not be mixed with other medicinal products or diluents.
- Visually inspect the diluted medicinal product for particles or discolouration prior to administration. Do not use if it is discoloured or if foreign particles are seen.
- The diluted medicinal product is to be used as soon as possible and within 24 hours of dilution. If the diluted medicinal product is stored at 2 °C to 8 °C (do not freeze), allow the solution to warm to room temperature prior to infusion.
- The diluted solution is to be infused intravenously over 1 hour at a rate of approximately 2 mL per minute.

- After the infusion is complete, flush the intravenous line with sodium chloride 9 mg/mL (0.9%) solution for injection.
- Each vial is for single–use only.
- Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MARKETING AUTHORISATION HOLDER

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

8. MARKETING AUTHORISATION NUMBER

EU/1/06/346/001

9. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

Date of first authorisation: 27th June 2006
Date of latest renewal: 18th April 2016

10. DATE OF REVISION OF THE TEXT

Detailed information on this medicinal product is available on the website of the European Medicines Agency https://www.ema.europa.eu.

1. NAME OF THE MEDICINAL PRODUCT

Tysabri 150 mg solution for injection in pre-filled syringe

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each mL contains 150 mg of natalizumab.

Natalizumab is a recombinant humanised anti-α4-integrin antibody produced in a murine cell line by recombinant DNA technology.

Excipient with known effect

Each pre-filled syringe contains 0.4 mg polysorbate 80 in 1 ml solution for injection (see section 4.4 for further information).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection (injection)

Colourless to slightly yellow, slightly opalescent to opalescent solution, with a pH of 5.8 – 6.4 and an osmolality of 114 – 144 mOsm/kg.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Tysabri is indicated as single disease modifying therapy in adults with highly active relapsing remitting multiple sclerosis (RRMS) for the following patient groups:

- Patients with highly active disease despite a full and adequate course of treatment with at least one disease modifying therapy (DMT) (for exceptions and information about washout periods see sections 4.4 and 5.1)

or

- Patients with rapidly evolving severe RRMS defined by 2 or more disabling relapses in one year, and with 1 or more Gadolinium enhancing lesions on brain Magnetic Resonance Imaging (MRI) or a significant increase in T2 lesion load as compared to a previous recent MRI.

4.2 Posology and method of administration

Therapy is to be initiated and continuously supervised by specialised physicians experienced in the diagnosis and treatment of neurological conditions, with timely access to MRI. Patients must be monitored for early signs and symptoms of Progressive Multifocal Leukoencephalopathy (PML). Patients treated with this medicinal product must be given the Patient Card and be informed about the risks of the medicinal product (see also package leaflet).
In case of administration by a healthcare professional outside a clinical setting, self-administration or administration by a caregiver (see below), the Pre-Administration Checklist should be provided (see section 4.4 for educational guidance).

After 2 years of treatment, patients should be re-informed about the risks, especially the increased risk of PML, and should be instructed together with their caregivers on early signs and symptoms of PML.

Resources for the management of hypersensitivity reactions and access to MRI should be available. There are limited data for the subcutaneous formulation in the Tysabri naïve patient population (see section 4.4).

Some patients may have been exposed to immunosuppressive medicinal products (e.g. mitoxantrone, cyclophosphamide, azathioprine). These medicinal products have the potential to cause prolonged immunosuppression, even after dosing is discontinued. Therefore, the physician must confirm that such patients are not immunocompromised before starting treatment (see section 4.4).

Posology

The recommended dose for subcutaneous administration is 300 mg every 4 weeks. As each pre-filled syringe contains 150 mg natalizumab, two pre-filled syringes need to be administered.

Continued therapy must be carefully reconsidered in patients who show no evidence of therapeutic benefit beyond 6 months.

Data on the safety and efficacy of natalizumab (intravenous infusion) at 2 years were generated from controlled, double–blind studies. After 2 years continued therapy should be considered only following a reassessment of the potential for benefit and risk. Patients should be re-informed about the risk factors for PML, like duration of treatment, immunosuppressant use prior to receiving the medicinal product and the presence of anti-John Cunningham virus (JCV) antibodies (see section 4.4).

Readministration

The efficacy of re-administration has not been established (for safety, see section 4.4).

Any switch in route of administration of the medicinal product should be made 4 weeks after the previous dose.

Special populations

Elderly

This medicinal product is not recommended for use in patients aged over 65 years due to a lack of data in this population.

Renal and hepatic impairment

Studies have not been conducted to examine the effects of renal or hepatic impairment.

The mechanism for elimination and results from population pharmacokinetics suggest that dose adjustment would not be necessary in patients with renal or hepatic impairment.

Paediatric population

The safety and efficacy of this medicinal product in children and adolescents up to 18 years have not been established. Currently available data are described in sections 4.8 and 5.1.

Method of administration

Tysabri 150 mg solution for injection in pre-filled syringe is for subcutaneous (SC) injection only. It is not intended for intravenous (IV) infusion.

Two pre-filled syringes should be administered (total dose 300 mg), one after the other without significant delay. The second injection should be administered no later than 30 minutes after the first injection.

The sites for subcutaneous injection are the thigh, abdomen (at least 6 cm away from the navel), or the posterior aspect of the upper arm (the latter only in case of injection by a healthcare professional or a caregiver). The injection should not be made into an area of the body where the skin is irritated, reddened, bruised, infected, or scarred in any way. When removing the syringe from the injection site, the plunger should be let go of while pulling the needle straight out. Letting go of the plunger will allow the needle guard to cover the needle. The second injection should be more than 3 cm away from the first injection location (see instructions for administration at the end of the package leaflet).

Natalizumab naïve patients should be observed during the injection and for 1 hour after for signs and symptoms of injection reactions including hypersensitivity for the first six natalizumab doses. For patients currently receiving natalizumab and who have already received at least six doses, regardless of the route of natalizumab administration used for the first six doses, the 1-hour post-injection observation time for subsequent subcutaneous injections may be reduced or removed according to clinical judgement if the patients have not experienced any injection/infusion reactions.

Administration outside the clinical setting (OCS)

Natalizumab injections administered by a healthcare professional outside a clinical setting (e.g. at home) may be considered for patients who have previously tolerated at least six doses of natalizumab well, i.e. who have not experienced hypersensitivity reactions. The decision for a patient to receive injections outside a clinical setting should be made after evaluation and recommendation by the specialised physician. Healthcare professionals should be vigilant for the early signs and symptoms of PML (see section 4.4 for further information on PML and educational guidance).

Self-administration or administration by a caregiver

Self-administration by the patient or administration by a caregiver may be considered for patients who have previously tolerated at least six doses of natalizumab well, i.e. who have not experienced hypersensitivity reactions. The decision should be made after evaluation and recommendation by the specialised physician.

Patients or caregivers must administer at least two doses via SC route (two injections each) under the guidance of a healthcare professional. They must be instructed to read the Patient Card and review the Pre-Administration Checklist prior to each dose. Patients or caregivers must be advised to remain vigilant for the early signs and symptoms of PML (see section 4.4 for further information on PML and educational guidance) and, if a hypersensitivity reaction occurs, to stop administration and seek medical attention immediately.

After a treatment gap of 3 months or more, the six subsequent doses must be administered under the supervision of a healthcare professional because of the concern for hypersensitivity reaction.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

Progressive multifocal leukoencephalopathy (PML).

Patients with increased risk for opportunistic infections, including immunocompromised patients (including those currently receiving immunosuppressive therapies or those immunocompromised by prior therapies (see sections 4.4 and 4.8)).

Combination with other DMTs.

Known active malignancies, except for patients with cutaneous basal cell carcinoma.

4.4 Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and batch number of the administered product should be clearly recorded.

Progressive Multifocal Leukoencephalopathy (PML)

Use of this medicinal product has been associated with an increased risk of PML, an opportunistic infection caused by JC virus, which may be fatal or result in severe disability. Due to this increased risk of developing PML, the benefits and risks of treatment should be individually reconsidered by the specialised physician and the patient; patients must be monitored at regular intervals throughout and should be instructed together with their caregivers on early signs and symptoms of PML. JC virus also causes JCV granule cell neuronopathy (GCN) which has been reported in patients treated with this medicinal product. Symptoms of JCV GCN are similar to symptoms of PML (i.e. cerebellar syndrome).

The following risk factors are associated with an increased risk of PML:

- The presence of anti-JCV antibodies.
- Treatment duration, especially beyond 2 years. After 2 years all patients should be re-informed about the risk of PML with the medicinal product.
- Immunosuppressant use prior to receiving the medicinal product.

Patients who are anti-JCV antibody positive are at an increased risk of developing PML compared to patients who are anti-JCV antibody negative. Patients who have all three risk factors for PML (i.e., are anti-JCV antibody positive **and** have received more than 2 years of therapy with this medicinal product **and** have received prior immunosuppressant therapy) have a significantly higher risk of PML.

In anti-JCV antibody positive natalizumab treated patients who have not used prior immunosuppressants the level of anti-JCV antibody response (index) is associated with the level of risk for PML.

In anti-JCV antibody positive patients, extended interval dosing of natalizumab (average dosing interval of approximately 6 weeks) is suggested to be associated with a lower PML risk compared to approved dosing. If utilising extended interval dosing, caution is required because the efficacy of extended interval dosing has not been established and the associated benefit/risk balance is currently unknown (see section 5.1). The decrease in PML risk is based on data from intravenous route of administration. No clinical data are available on either the safety or efficacy of this extended interval dosing with subcutaneous route of administration. For further information, refer to the Healthcare Professional Guide.

Patients considered at high risk with this treatment should only continue the treatment if the benefits outweigh the risks. For the estimation of PML risk in the different patient subgroups, please refer to the Healthcare Professional Guide.

Anti-JCV antibody testing

Anti-JCV antibody testing provides supportive information for risk stratification of treatment with this medicinal product. Testing for serum anti-JCV antibody prior to initiating therapy or in patients receiving the medicinal product with an unknown antibody status is recommended. Anti-JCV antibody negative patients may still be at risk of PML for reasons such as a new JCV infection, fluctuating antibody status or a false negative test result. Re-testing of anti-JCV antibody negative patients every 6

months is recommended. Retesting low index patients who have no history of prior immunosuppressant use every 6 months once they reach the 2 year treatment point is recommended.

The anti-JCV antibody assay (ELISA) should not be used to diagnose PML. Use of plasmapheresis/plasma exchange (PLEX) or intravenous immunoglobulin (IVIg) can affect meaningful interpretation of serum anti-JCV antibody testing. Patients should not be tested for anti-JCV antibodies within 2 weeks of PLEX due to removal of antibodies from the serum, or within 6 months of IVIg (i.e. 6 months = 5 half-lives for immunoglobulins).

For further information on anti-JCV antibody testing please see the Healthcare Professional Guide.

MRI screening for PML

Before initiation of treatment with this medicinal product, a recent (usually within 3 months) MRI should be available as a reference and be repeated at least on a yearly basis. More frequent MRIs (e.g. on a 3 to 6 monthly basis) using an abbreviated protocol should be considered for patients at higher risk of PML. This includes:

- Patients who have all three risk factors for PML (i.e., are-anti-JCV antibody positive **and** have received more than 2 years of therapy with this medicinal product **and** have received prior immunosuppressant therapy),

or

- Patients with a high anti-JCV antibody index who have received more than 2 years of therapy with this medicinal product and without prior history of immunosuppressant therapy.

Current evidence suggests that the risk of PML is low at an index equal to or below 0.9 and increases substantially above 1.5 for patients who have been on treatment with this medicinal product for longer than 2 years (see the Healthcare Professional Guide for further information).

PML should be considered as a differential diagnosis in any MS patient taking natalizumab presenting with neurological symptoms and/or new brain lesions in MRI. Cases of asymptomatic PML based on MRI and positive JCV DNA in the cerebrospinal fluid have been reported.

Physicians should refer to the Healthcare Professional Guide for further information on managing the risk of PML in natalizumab-treated patients.

If PML or JCV GCN is suspected, further dosing must be suspended until PML has been excluded.

The specialised physician should evaluate the patient to determine if the symptoms are indicative of neurological dysfunction and, if so, whether these symptoms are typical of MS or possibly suggestive of PML or JCV GCN. If any doubt exists, further evaluation, including MRI scan preferably with contrast (compared with pre-treatment baseline MRI), CSF testing for JC Viral DNA and repeat neurological assessments, should be considered as described in the Healthcare Professional Guide (see Educational guidance). Once the physician has excluded PML and/or JCV GCN (if necessary, by repeating clinical, imaging and/or laboratory investigations if clinical suspicion remains), dosing may resume.

The physician should be particularly alert to symptoms suggestive of PML or JCV GCN that the patient may not notice (e.g. cognitive, psychiatric symptoms or cerebellar syndrome). Patients should also be advised to inform their partner or caregivers about their treatment, since they may notice symptoms that the patient is not aware of.

PML has been reported following discontinuation of this medicinal product in patients who did not have findings suggestive of PML at the time of discontinuation. Patients and physicians should continue to follow the same monitoring protocol and be alert for any new signs or symptoms that may be suggestive of PML for approximately 6 months following discontinuation of natalizumab.

If a patient develops PML the dosing of this medicinal product must be permanently discontinued.

Following reconstitution of the immune system in immunocompromised patients with PML improved outcome has been seen.

Based on a retrospective analysis of natalizumab-treated patients, no difference was observed on 2-year survival after PML diagnosis between patients who received PLEX and those who did not. For other considerations on the management of PML, see the Healthcare Professional Guide.

PML and IRIS (Immune Reconstitution Inflammatory Syndrome)

IRIS occurs in almost all PML patients treated with this medicinal product after withdrawal or removal of the medicinal product. IRIS is thought to result from the restoration of immune function in patients with PML, which can lead to serious neurological complications and may be fatal. Monitoring for development of IRIS and appropriate treatment of the associated inflammation during recovery from PML should be undertaken (see the Healthcare Professional Guide for further information).

Infections including other opportunistic infections

Other opportunistic infections have been reported with use of this medicinal product, primarily in patients with Crohn's disease who were immunocompromised or where significant comorbidity existed, however increased risk of other opportunistic infections with use of the medicinal product in patients without these co-morbidities cannot currently be excluded. Opportunistic infections were also detected in MS patients treated with this medicinal product as a monotherapy (see section 4.8).

This treatment increases the risk of developing encephalitis and meningitis caused by herpes simplex and varicella zoster viruses. Serious, life-threatening, and sometimes fatal cases have been reported in the post-marketing setting in multiple sclerosis patients receiving the treatment (see section 4.8). If herpes encephalitis or meningitis occurs, the medicinal product should be discontinued, and appropriate treatment for herpes encephalitis or meningitis should be administered.

Acute retinal necrosis (ARN) is a rare fulminant viral infection of the retina caused by the family of herpes viruses (e.g. varicella zoster). ARN has been observed in patients being administered this medicinal product and can be potentially blinding. Patients presenting with eye symptoms such as decreased visual acuity, redness and painful eye should be referred for retinal screening for ARN. Following clinical diagnosis of ARN, discontinuation of this medicinal product should be considered in these patients.

Prescribers should be aware of the possibility that other opportunistic infections may occur during therapy and should include them in the differential diagnosis of infections that occur in Tysabri-treated patients. If an opportunistic infection is suspected, dosing is to be suspended until such infections can be excluded through further evaluations.

If a patient receiving this medicinal product develops an opportunistic infection, dosing of the medicinal product must be permanently discontinued.

Educational guidance

All physicians who intend to prescribe the medicinal product must ensure they are familiar with the Healthcare Professional Guide.

Physicians must discuss the benefits and risks of natalizumab therapy with the patient and provide them with a Patient Card. Patients should be instructed that if they develop any infection then they should inform their physician that they are being treated with this medicinal product.

Physicians should counsel patients on the importance of uninterrupted dosing, particularly in the early months of treatment (see Hypersensitivity).

Healthcare professionals administering natalizumab subcutaneous injection outside a clinical setting, e.g. at home, must review the Pre-Administration Checklist for each patient prior to each administration. In case of administration by the patient or by a caregiver, they must be instructed to review the Pre-Administration Checklist prior to each dose.

Hypersensitivity

Hypersensitivity reactions have been associated with this medicinal product, including, for the intravenous infusion, serious systemic reactions (see section 4.8).

These reactions usually occurred within one hour after administration. The risk for hypersensitivity was greatest with early infusions and in patients re-exposed to treatment following an initial short exposure (one or two infusions) and extended period (3 months or more) without treatment. However, the risk of hypersensitivity reactions should be considered for every administration.

Patients are to be observed during the subcutaneous injections and for 1 hour after, for signs and symptoms of injection reactions including hypersensitivity (see sections 4.2 and 4.8). Resources for the management of hypersensitivity reactions should be available. In case of administration by the patient or a caregiver, they should be informed of the signs and symptoms of hypersensitivity reactions. If a hypersensitivity reaction occurs, patients or caregivers should be advised to stop administration and seek medical attention immediately.

This medicinal product should be discontinued and appropriate therapy initiated at the first symptoms or signs of hypersensitivity.

Patients who have experienced a hypersensitivity reaction must be permanently discontinued from treatment with natalizumab.

There are limited data for the subcutaneous formulation in the Tysabri naïve patient population (see section 5.1).

Concurrent treatment with immunosuppressants

The safety and efficacy of this medicinal product in combination with other immunosuppressive and antineoplastic therapies have not been fully established. Concurrent use of these agents with this medicinal product may increase the risk of infections, including opportunistic infections, and is contraindicated (see section 4.3).

In phase 3 MS clinical trials with natalizumab intravenous infusion, concomitant treatment of relapses with a short course of corticosteroids was not associated with an increased rate of infection. Short courses of corticosteroids can be used in combination with this medicinal product.

Prior treatment with immunosuppressive or immunomodulatory therapies

Patients with a treatment history of immunosuppressant medications are at increased risk for PML.

Data from an observational study demonstrated that there is no increased risk of PML for the group of patients switching to natalizumab from fingolimod, dimethyl fumarate, or teriflunomide when compared to the group of patients switching from either beta interferon or glatiramer acetate.

No studies have been performed to evaluate the safety of natalizumab when switching patients from DMTs other than beta interferon, glatiramer acetate, fingolimod, dimethyl fumarate and teriflunomide. It is unknown if patients switching from other therapies to natalizumab have an increased risk of PML compared to those switching from the above-mentioned DMTs, therefore these patients should be monitored more frequently (i.e. similarly to patients switching from immunosuppressants to natalizumab).

Care should be taken with patients who have previously received immunosuppressants to allow sufficient time for immune function recovery to occur. Physicians must evaluate each individual case to determine whether there is evidence of an immunocompromised state prior to commencing treatment (see section 4.3).

When switching patients from another DMT to this medicinal product, the half-life and mode of action of the other therapy must be considered to avoid an additive immune effect whilst at the same time minimising the risk of disease reactivation. A Complete Blood Count (CBC, including lymphocytes) is recommended prior to initiating treatment to ensure that immune effects of the previous therapy (i.e. cytopenia) have resolved.

Patients can switch directly from beta interferon or glatiramer acetate to natalizumab providing there are no signs of relevant treatment-related abnormalities e.g. neutropenia and lymphopenia.

When switching from dimethyl fumarate, the washout period should be sufficient for lymphocyte count to recover before treatment is started.

Following discontinuation of fingolimod, lymphocyte count progressively returns to normal range within 1 to 2 months after stopping therapy. The washout period should be sufficient for lymphocyte count to recover before treatment is started.

Teriflunomide is eliminated slowly from the plasma. Without an accelerated elimination procedure, clearance of teriflunomide from plasma can take from several months up to 2 years. An accelerated elimination procedure as defined in the teriflunomide Summary of Product Characteristics is recommended or alternatively washout period should not be shorter than 3.5 months. Caution regarding potential concomitant immune effects is required when switching patients from teriflunomide to this medicinal product.

Alemtuzumab has profound prolonged immunosuppressive effects. As the actual duration of these effects is unknown, initiating treatment with this medicinal product after alemtuzumab is not recommended unless the benefits clearly outweigh the risks for the individual patient.

Immunogenicity

Disease exacerbations or injection related events may indicate the development of antibodies against natalizumab. In these cases, the presence of antibodies should be evaluated and if these remain positive in a confirmatory test after at least 6 weeks, treatment should be discontinued, as persistent antibodies are associated with a substantial decrease in efficacy of this medicinal product and an increased incidence of hypersensitivity reactions (see section 4.8).

Since patients who have received an initial short exposure to this medicinal product and then had an extended period without treatment are at a higher risk of developing anti-natalizumab antibodies and/or hypersensitivity upon redosing, the presence of antibodies should be evaluated and if these remain positive in a confirmatory test after at least 6 weeks, the patient should not receive further treatment with natalizumab (see section 5.1).

Hepatic events

Spontaneous serious adverse reactions of liver injury have been reported during the post-marketing phase (see section 4.8). These liver injuries may occur at any time during treatment, even after the first

dose. In some instances, the reaction reoccurred when treatment was reintroduced. Some patients with a past medical history of an abnormal liver test have experienced an exacerbation of abnormal liver test while on treatment. Patients should be monitored as appropriate for impaired liver function and be instructed to contact their physician in case signs and symptoms suggestive of liver injury occur, such as jaundice and vomiting. In cases of significant liver injury this medicinal product should be discontinued.

Thrombocytopenia

Thrombocytopenia, including immune thrombocytopenic purpura (ITP), has been reported with the use of natalizumab. Delay in the diagnosis and treatment of thrombocytopenia may lead to serious and life-threatening sequelae. Patients should be instructed to report to their physician immediately if they experience any signs of unusual or prolonged bleeding, petechiae, or spontaneous bruising. If thrombocytopenia is identified, discontinuation of natalizumab should be considered.

Stopping therapy

If a decision is made to stop treatment with natalizumab, the physician needs to be aware that natalizumab remains in the blood and has pharmacodynamic effects (e.g increased lymphocyte counts) for approximately 12 weeks following the last dose. Starting other therapies during this interval will result in a concomitant exposure to natalizumab. For medicinal products such as interferon and glatiramer acetate, concomitant exposure of this duration was not associated with safety risks in clinical trials. No data are available in MS patients regarding concomitant exposure with immunosuppressant medication. Use of these medicinal products soon after the discontinuation of natalizumab may lead to an additive immunosuppressive effect. This should be carefully considered on a case-by-case basis, and a wash-out period of natalizumab might be appropriate. Short courses of steroids used to treat relapses were not associated with increased infections in clinical trials.

Polysorbate 80 (E 433) content

This medicinal product contains 0.4 mg of polysorbate 80 in each pre-filled syringe, which is equivalent to 0.8 mg per dose. Polysorbates may cause allergic reactions.

Sodium content

This medicinal product contains less than 1 mmol sodium (23 mg) per dose (300 mg natalizumab), that is to say essentially ‘sodium-free’.

4.5 Interaction with other medicinal products and other forms of interaction

Natalizumab is contraindicated in combination with other DMTs (see section 4.3).

Immunisations

In a randomised, open label study of 60 patients with relapsing MS there was no significant difference in the humoral immune response to a recall antigen (tetanus toxoid) and only slightly slower and reduced humoral immune response to a neoantigen (keyhole limpet haemocyanin) was observed in patients who were treated with this medicinal product for 6 months compared to an untreated control group. Live vaccines have not been studied.

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

If a woman becomes pregnant while taking this medicinal product, discontinuation of treatment should be considered. A benefit/risk evaluation of the use of this medicinal product during pregnancy should

take into account the patient's clinical condition and the possible return of disease activity after stopping the medicinal product.

Pregnancy

Studies in animals have shown reproductive toxicity (see section 5.3).

Data from clinical trials, a prospective pregnancy registry, post-marketing cases and available literature do not suggest an effect of this medicinal product exposure on pregnancy outcomes.

The completed prospective Tysabri pregnancy registry contained 355 pregnancies with available outcomes. There were 316 live births, 29 of which were reported to have birth defects. Sixteen of the 29 were classified as major defects. The rate of defects corresponds to the defect rates reported in other pregnancy registries involving MS patients. There is no evidence of a specific pattern of birth defects with this medicinal product.

There are no adequate and well-controlled studies of natalizumab therapy in pregnant women.

Thrombocytopenia and anaemia in infants born to women exposed to natalizumab during pregnancy were reported in the post-marketing setting. Monitoring of platelet counts, haemoglobin, and haematocrit is recommended in neonates born to women exposed to natalizumab during pregnancy.

This medicinal product should be used during pregnancy only if clearly needed. If a woman becomes pregnant while taking natalizumab, discontinuation of natalizumab should be considered.

Breast-feeding

Natalizumab is excreted in human milk. The effect of natalizumab on newborn/infants is unknown. Breast-feeding should be discontinued during treatment with natalizumab.

Fertility

Reductions in female guinea pig fertility were observed in one study at doses in excess of the human dose; natalizumab did not affect male fertility. It is considered unlikely that natalizumab will affect fertility performance in humans following the maximum recommended dose.

4.7 Effects on ability to drive and use machines

Tysabri has a minor influence on the ability to drive and use machines. Dizziness may occur following administration of natalizumab (see section 4.8).

4.8 Undesirable effects

Summary of the safety profile

The safety profile observed for natalizumab administered subcutaneously was consistent with the known safety profile of natalizumab administered intravenously, with the exception of injection site pain. The overall frequency of injection site pain was common 4% (3/71) for subjects receiving natalizumab 300 mg, every 4 weeks, by subcutaneous administration.

In placebo-controlled trials in 1 617 MS patients treated with natalizumab (intravenous infusion), for up to 2 years (placebo: 1 135), adverse events leading to discontinuation of therapy occurred in 5.8% of patients treated with natalizumab (placebo: 4.8%). Over the 2-year duration of the studies, 43.5% of patients treated with natalizumab reported adverse reactions (placebo: 39.6%).

In clinical trials in 6 786 patients treated with natalizumab (intravenous infusion and subcutaneous injection), the most frequently occurring adverse reactions were headache (32%), nasopharyngitis

(27%), fatigue (23%), urinary tract infection (16%), nausea (15%), arthralgia (14%), and dizziness (11%) associated with natalizumab administration.

Tabulated list of adverse reactions

Adverse reactions arising from clinical studies, post-authorisation safety studies and spontaneous reports are presented in Table 1 below. Within the system organ classes they are listed under the following headings: Very common (≥1/10); Common (≥1/100 to <1/10); Uncommon (≥1/1 000 to <1/100); Rare (≥1/10 000 to <1/1 000); Very rare (<1/10 000), not known (cannot be estimated from the available data). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1. Adverse reactions

MedDRA System Organ Class	Frequency of adverse reactions				
	Very Common	*Common*	*Uncommon*	*Rare*	*Not known*
Infections and infestations	Nasopharyngitis Urinary tract infection	Herpes infection	Progressive multifocal leukoencephalopathy	Herpes ophthalmic	Meningoencephalitis herpetic JC virus granule cell neuropathy Necrotising herpetic retinopathy
Blood and lymphatic system disorders		Anaemia	Thrombocytopenia, Immune thrombocytopenic purpura (ITP), Eosinophilia	Haemolytic anaemia Nucleated red cells	
Immune system disorders		Hypersensitivity	Anaphylactic reaction Immune reconstitution inflammatory syndrome		
Nervous system disorders	Dizziness Headache				
Vascular disorders		Flushing			
Respiratory, thoracic and mediastinal disorders		Dyspnoea			
Gastrointestinal disorders	Nausea	Vomiting			
Hepatobiliary disorders				Hyperbilirubinaemia	Liver injury
Skin and subcutaneous tissue disorders		Pruritus Rash Urticaria		Angioedema	
Musculoskeletal and connective tissue disorders	Arthralgia				
General disorders and administration site conditions	Fatigue	Pyrexia Chills Infusion site reaction Injection site reaction	Face oedema		

MedDRA System Organ Class	Frequency of adverse reactions				
	Very Common	*Common*	*Uncommon*	*Rare*	*Not known*
Investigations		Hepatic enzyme increased Drug specific antibody present			
Injury, poisoning and procedural complications	Infusion related reaction				

Description of selected adverse reactions

Hypersensitivity reactions

Hypersensitivity reactions usually occurred within one hour after completion of the subcutaneous injections. The number of patients analysed in the DELIVER and REFINE studies was low (see section 5.1).

In 2-year controlled clinical trials in MS patients receiving natalizumab intravenously, hypersensitivity reactions occurred in up to 4% of patients. Anaphylactic/anaphylactoid reactions occurred in less than 1% of patients receiving this medicinal product. Hypersensitivity reactions usually occurred during the infusion or within the 1-hour period after the completion of the infusion (see section 4.4). In post-marketing experience, there have been reports of hypersensitivity reactions which have occurred with one or more of the following associated symptoms: hypotension, hypertension, chest pain, chest discomfort, dyspnoea, angioedema, in addition to more usual symptoms such as rash and urticaria.

Immunogenicity

In 10% of patients, antibodies against natalizumab were detected in 2-year controlled clinical trials in MS patients receiving natalizumab intravenously. Persistent anti-natalizumab antibodies (one positive test reproducible on retesting at least 6 weeks later) developed in approximately 6% of patients. Antibodies were detected on only one occasion in an additional 4% of patients. Persistent antibodies were associated with a substantial decrease in the effectiveness of natalizumab and an increased incidence of hypersensitivity reactions. Additional infusion-related reactions associated with persistent antibodies included rigors, nausea, vomiting and flushing (see section 4.4). In the 32-week DELIVER study in MS patients with no prior exposure to natalizumab, persistent anti-natalizumab antibodies developed in 1 subject (4%) from 26 subjects who received natalizumab subcutaneously. Antibodies were detected on only one occasion in another 5 subjects (19%). In the 60-week REFINE study in MS patients, no subjects (136 subjects) who switched from natalizumab intravenous administration to subcutaneous administration had detectable ADA during the study (see section 5.1).

If, after approximately 6 months of therapy, persistent antibodies are suspected, either due to reduced efficacy or due to occurrence of infusion-related events, they may be detected and confirmed with a subsequent test 6 weeks after the first positive test. Given that efficacy may be reduced, or the incidence of hypersensitivity or infusion-related reactions may be increased in a patient with persistent antibodies, treatment should be discontinued in patients who develop persistent antibodies.

Infections, including PML and opportunistic infections

In 2-year controlled clinical trials in MS patients, the rate of infection was approximately 1.5 per patient-year in both natalizumab (intravenously)- and placebo-treated patients. The nature of the infections was generally similar in natalizumab- and placebo-treated patients. A case of *cryptosporidium* diarrhoea was reported in MS clinical trials. In other clinical trials, cases of additional

opportunistic infections have been reported, some of which were fatal. The majority of patients did not interrupt natalizumab therapy during infections and recovery occurred with appropriate treatment.

In clinical trials (intravenous formulation), herpes infections (Varicella-Zoster virus, Herpes-simplex virus) occurred slightly more frequently in natalizumab-treated patients than in placebo-treated patients. In post-marketing experience, serious, life-threatening, and sometimes fatal cases of encephalitis and meningitis caused by herpes simplex or varicella zoster have been reported in multiple sclerosis patients receiving natalizumab. The duration of treatment with natalizumab prior to onset ranged from a few months to several years (see section 4.4).

In post-marketing experience, rare cases of ARN have been observed in patients receiving this medicinal product. Some cases have occurred in patients with central nervous system (CNS) herpes infections (e.g. herpes meningitis and encephalitis). Serious cases of ARN, either affecting one or both eyes, led to blindness in some patients. The treatment reported in these cases included anti-viral therapy and, in some cases, surgery (see section 4.4).

Cases of PML have been reported from clinical trials, post-marketing observational studies and post-marketing passive surveillance. PML usually leads to severe disability or death (see section 4.4). Cases of JCV GCN have also been reported during post-marketing use of this medicinal product Symptoms of JCV GCN are similar to PML.

Hepatic events

Spontaneous cases of serious liver injuries, increased liver enzymes, hyperbilirubinaemia have been reported during the post-marketing phase (see section 4.4).

Anaemia and haemolytic anaemia

Rare, serious cases of anaemia and haemolytic anaemia have been reported in patients treated with natalizumab in post-marketing observational studies.

Effects on laboratory tests

In 2-year controlled clinical trials in MS patients, treatment with natalizumab was associated with increases in circulating lymphocytes, monocytes, eosinophils, basophils and nucleated red blood cells. Elevations in neutrophils were not seen. Increases from baseline for lymphocytes, monocytes, eosinophils and basophils ranged from 35% to 140% for individual cell types but mean cell counts remained within normal ranges with intravenous infusion administration. During treatment with this medicinal product, small reductions in haemoglobin (mean decrease 0.6 g/dL), haematocrit (mean decrease 2%) and red blood cell counts (mean decrease 0.1 x 10^6/L) were seen. All changes in haematological variables returned to pre-treatment values, usually within 16 weeks of last dose of the medicinal product and the changes were not associated with clinical symptoms. In post-marketing experience, there have also been reports of eosinophilia (eosinophil count > 1 500/mm^3) without clinical symptoms. In such cases where therapy was discontinued the elevated eosinophil levels resolved.

Thrombocytopenia
In post-marketing experience, thrombocytopenia and immune thrombocytopenic purpura (ITP) have been reported with uncommon frequency.

Paediatric population

Serious adverse events were evaluated in 621 MS paediatric patients included in a meta-analysis (see also section 5.1). Within the limitations of these data, there were no new safety signals identified in this patient population. 1 case of herpes meningitis was reported in the meta-analysis. No cases of PML were identified in the meta-analysis, however, PML has been reported in natalizumab-treated paediatric patients in the post-marketing setting.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system listed in Appendix V.

4.9 Overdose

Safety of doses higher than 300 mg has not been adequately evaluated. The maximum amount of natalizumab that can be safely administered has not been determined.

There is no known antidote for natalizumab overdose. Treatment consists of discontinuation of the medicinal product and supportive therapy as needed.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Immunosuppressants, monoclonal antibodies, ATC code: L04AG03

Pharmacodynamic effects

Natalizumab is a selective adhesion-molecule inhibitor and binds to the α4-subunit of human integrins, which is highly expressed on the surface of all leukocytes, with the exception of neutrophils. Specifically, natalizumab binds to the α4β1 integrin, blocking the interaction with its cognate receptor, vascular cell adhesion molecule-1 (VCAM-1), and ligands osteopontin, and an alternatively spliced domain of fibronectin, connecting segment-1 (CS-1). Natalizumab blocks the interaction of α4β7 integrin with the mucosal addressin cell adhesion molecule-1 (MadCAM-1). Disruption of these molecular interactions prevents transmigration of mononuclear leukocytes across the endothelium into inflamed parenchymal tissue. A further mechanism of action of natalizumab may be to suppress ongoing inflammatory reactions in diseased tissues by inhibiting the interaction of α4-expressing leukocytes with their ligands in the extracellular matrix and on parenchymal cells. As such, natalizumab may act to suppress inflammatory activity present at the disease site, and inhibit further recruitment of immune cells into inflamed tissues.

In MS, lesions are believed to occur when activated T-lymphocytes cross the blood-brain barrier (BBB). Leukocyte migration across the BBB involves interaction between adhesion molecules on inflammatory cells and endothelial cells of the vessel wall. The interaction between α4β1 and its targets is an important component of pathological inflammation in the brain and disruption of these interactions leads to reduced inflammation. Under normal conditions, VCAM-1 is not expressed in the brain parenchyma. However, in the presence of pro-inflammatory cytokines, VCAM-1 is upregulated on endothelial cells and possibly on glial cells near the sites of inflammation. In the setting of central nervous system (CNS) inflammation in MS, it is the interaction of α4β1 with VCAM-1, CS-1 and osteopontin that mediates the firm adhesion and transmigration of leukocytes into the brain parenchyma and may perpetuate the inflammatory cascade in CNS tissue. Blockade of the molecular interactions of α4β1 with its targets reduces inflammatory activity present in the brain in MS and inhibits further recruitment of immune cells into inflamed tissue, thus reducing the formation or enlargement of MS lesions.

The EC50 of natalizumab binding to α4β1 integrin is estimated to be 2.04 mg/L based on a population pharmacokinetic/pharmacodynamic model. There was no difference in α4β1integrin binding after natalizumab 300 mg every 4 weeks was administered subcutaneously or intravenously. Mean PD (alpha-4 saturation on mononuclear lymphocyte cells) was similar between the intravenous

administration Q6W and Q4W regimens, with the difference in mean percent alpha-4 saturation ranging from 9 to 16%.

Clinical efficacy

Based on similarities in pharmacokinetics and pharmacodynamics between intravenous and subcutaneous administration, the efficacy data from intravenous infusion is provided as well as those from patients receiving the subcutaneous injection.

AFFIRM clinical study

Efficacy as monotherapy for intravenous infusion has been evaluated in one randomised, double-blind, placebo-controlled study lasting 2 years (AFFIRM study) in RRMS patients who had experienced at least 1 clinical relapse during the year prior to entry and had a Kurtzke Expanded Disability Status Scale (EDSS) score between 0 and 5. Median age was 37 years, with a median disease duration of 5 years. The patients were randomised with a 2:1 ratio to receive natalizumab 300 mg (n = 627) or placebo (n = 315) every 4 weeks for up to 30 infusions. Neurological evaluations were performed every 12 weeks and at times of suspected relapse. MRI evaluations for T1-weighted gadolinium (Gd)-enhancing lesions and T2-hyperintense lesions were performed annually.

Study features and results are presented in the Table 2.

Table 2. AFFIRM study: Main features and results		
Design	Monotherapy; randomised double-blind placebo-controlled parallel-group trial for 120 weeks	
Subjects	RRMS (McDonald criteria)	
Treatment	Placebo / Natalizumab 300 mg i.v. every 4 weeks	
One year endpoint	Relapse rate	
Two year endpoint	Progression on EDSS	
Secondary endpoints	Relapse rate derived variables / MRI-derived variables	
Subjects	Placebo	Natalizumab
Randomised	315	627
Completing 1 years	296	609
Completing 2 years	285	589
Age yrs, median (range)	37 (19-50)	36 (18-50)
MS-history yrs, median (range)	6.0 (0-33)	5.0 (0-34)
Time since diagnosis, yrs median (range)	2.0 (0-23)	2.0 (0-24)
Relapses in previous 12 months, median (range)	1.0 (0-5)	1.0 (0-12)
EDSS-baseline, median (range)	2 (0-6.0)	2 (0-6.0)
RESULTS		
Annual relapse rate		
After one year (primary endpoint)	0.805	0.261
After two years	0.733	0.235
One year	Rate ratio 0.33 $CI_{95\%}$ 0.26 ; 0.41	
Two years	Rate ratio 0.32 $CI_{95\%}$ 0.26 ; 0.40	
Relapse free		
After one year	53%	76%
After two years	41%	67%
Disability		
Proportion progressed[1](12-week confirmation; primary outcome)	29%	17%

Table 2. AFFIRM study: Main features and results		
Design	Monotherapy; randomised double-blind placebo-controlled parallel-group trial for 120 weeks	
	Hazard ratio 0.58, $CI_{95\%}$ 0.43; 0.73, p<0.001	
Proportion progressed[1](24-week confirmation)	23%	11%
	Hazard ratio 0.46, $CI_{95\%}$ 0.33; 0.64, p<0.001	
MRI (0-2 years)		
Median % change in T2-hyperintense lesion volume	+8.8%	-9.4% (p<0.001)
Mean number of new or newly-enlarging T2-hyperintense lesions	11.0	1.9 (p<0.001)
Mean number of T1-hypointense lesions	4.6	1.1 (p<0.001)
Mean number of Gd-enhancing lesions	1.2	0.1 (p<0.001)

[1] Progression of disability was defined as at least a 1.0 point increase on the EDSS from a baseline EDSS >= 1.0 sustained for 12 or 24 weeks or at least a 1.5 point increase on the EDSS from a baseline EDSS = 0 sustained for 12 or 24 weeks.

In the sub-group of patients indicated for treatment of rapidly evolving RRMS (patients with 2 or more relapses and 1 or more Gd+ lesion), the annualised relapse rate was 0.282 in the natalizumab-treated group (n = 148) and 1.455 in the placebo group (n = 61) (p < 0.001). Hazard ratio for disability progression was 0.36 (95% CI: 0.17, 0.76) p = 0.008. These results were obtained from a *post hoc* analysis and should be interpreted cautiously. No information on the severity of the relapses before inclusion of patients in the study is available.

Tysabri Observational Program (TOP, IMA-06-02)

The Tysabri Observational Program (TOP, IMA-06-02) was a multicentre, single-arm study initiated in 2007 to assess real-world safety and effectiveness outcomes in MS patients treated with Tysabri, with patient follow-up data for up to approximately 15 years. The study generated data from 6 319 patients in 17 countries, including 1 145 patients with up to 10 years of exposure and 102 patients with up to 15 years of exposure.

Overall, the safety findings of the TOP analyses were consistent with the known safety profile of natalizumab. Patients showed reductions in pre-treatment annualised relapse rate (ARR) regardless of the number of prior relapses, baseline EDSS, prior immunosuppressants use, or the number of DMTs used prior to natalizumab initiation. In the overall population, the ARR was 0.17 (95% CI: 0.17, 0.18) over the course of 15 years of follow up. Mean EDSS scores were similar from baseline (3.5, SD = 1.61) to year 15 (3.4, SD = 1.97) in patients treated with natalizumab.

A total of 5 635 patients had received another DMT prior to natalizumab initiation. The patients switching from beta interferon, glatiramer acetate or fingolimod had similar effectiveness outcomes to the overall natalizumab-treated population.

Paediatric population

A post-marketing meta-analysis was conducted using data from 621 paediatric MS patients treated with natalizumab (median age 17 years, range was 7 to 18 years, 91% aged ≥14 years). Within this analysis, a limited subset of patients with data available prior to treatment (158 of the 621 patients) demonstrated a reduction in ARR from 1.466 (95% CI 1.337, 1.604) prior to treatment to 0.110 (95% CI 0.094, 0.128).

Extended interval dosing

In a pre-specified, retrospective analysis of US anti-JCV antibody positive Tysabri patients intravenously administered (TOUCH Prescribing Program), the risk of PML was compared between

patients treated with the approved dosing interval and patients treated with extended interval dosing as identified in the last 18 months of exposure (EID, average dosing intervals of approximately 6 weeks). The majority (85%) of patients dosed with EID had received the approved dosing for ≥ 1 year prior to switching to EID. The analysis showed a lower risk of PML in patients treated with EID (hazard ratio = 0.06, 95% CI of hazard ratio = 0.01 to 0.22). The efficacy of this medicinal product when administered with EID has not been established, and therefore the benefit/risk balance of EID is unknown (see section 4.4).

Efficacy has been modelled for patients who switch to longer dosing after ≥ 1 year of approved dosing with this medicinal product under intravenous administration and who did not experience a relapse in the year prior to switching. Current pharmacokinetic/pharmacodynamic statistical modelling and simulation indicate that the risk of MS disease activity for patients switching to longer dosing intervals may be higher for patients with dosing intervals ≥ 7 weeks. No prospective clinical studies have been completed to validate these findings.

No clinical data are available on either the safety or efficacy of this extended interval dosing with the subcutaneous route of administration.

REFINE clinical study (subcutaneous formulation, population pre-treated with natalizumab [intravenous infusion] for a minimum of 12 months)

Subcutaneous administration was assessed in a randomised, blinded, parallel-group, phase 2 study (REFINE) exploring the safety, tolerability, and efficacy of multiple regimens of natalizumab (300 mg intravenous every 4 weeks, 300 mg subcutaneous every 4 weeks, 300 mg intravenous every 12 weeks, 300 mg subcutaneous every 12 weeks, 150 mg intravenous every 12 weeks and 150 mg subcutaneous every 12 weeks) in adult subjects (n = 290) with relapsing remitting multiple sclerosis conducted over a 60 week period. Subjects had received natalizumab for at least 12 months and were free from relapse for 12 months prior to randomisation. The primary objective of this study was to explore the effects of multiple regimens of natalizumab on disease activity and safety in subjects with RRMS. The primary endpoint of this study was the cumulative number of combined unique active (CUA) MRI lesions (sum of new Gd+ lesions on brain MRI and new or newly enlarging T2 hyperintense lesions not associated with Gd+ on T1 weighted scans). The mean CUA for the 300 mg subcutaneous every 4 week arm was low (0.02) and comparable to the 300 mg intravenous every 4 weeks arm (0.23). The CUA in the every 12 week treatment arms was significantly higher than the every 4 week treatment arms resulting in the early discontinuation of the every 12 week arms. Due to the exploratory nature of this study, no formal efficacy comparisons were made.

DELIVER clinical study (subcutaneous formulation, natalizumab naïve population)

The efficacy and safety of natalizumab for subcutaneous administration in the natalizumab naïve MS population was evaluated in a phase 1 randomised, open-label, dose-ranging study (DELIVER). Twelve subjects with RRMS and 14 subjects with secondary progressive MS were enrolled in the subcutaneous treatment arms. The study's primary objective was to compare the pharmacokinetics (PK) and pharmacodynamics (PD) of single subcutaneous or intramuscular 300-mg doses of natalizumab with intravenous infusion 300-mg doses of natalizumab in patients with multiple sclerosis (MS). Secondary objectives included investigation of the safety, tolerability, and immunogenicity of repeated subcutaneous and intramuscular natalizumab doses. An exploratory endpoint of this study included the number of new Gd+ lesions on brain MRI from baseline to Week 32. None of the subjects treated with natalizumab had any Gd+ lesions post-baseline, regardless of their disease stage (RRMS or secondary progressive MS), assigned route of administration, or the presence of Gd+ lesions at baseline. Across the RRMS and secondary progressive MS populations, 2 patients in the natalizumab 300 mg subcutaneous group experienced relapses compared to 3 patients in the natalizumab 300 mg intravenous infusion group. Small sample sizes and inter- and intra-patient variability prevent meaningful comparisons of efficacy data between groups.

5.2 Pharmacokinetic properties

The pharmacokinetics of natalizumab after subcutaneous administration was evaluated in 2 studies. DELIVER was a phase 1, randomised, open-label, dose-ranging study to evaluate the pharmacokinetics of subcutaneous and intramuscular natalizumab in subjects with MS (RRMS or secondary progressive MS) (n = 76). (see section 5.1 for a description of the REFINE study).

An updated population pharmacokinetic analysis was conducted consisting of 11 studies (conducted with subcutaneously and intravenously administered natalizumab) and data with serial PK sampling as measured by an industry standard assay. It included over 1 286 subjects receiving doses ranging from 1 to 6 mg/kg and fixed doses of 150/300 mg.

Absorption

The absorption from the injection site to systemic circulation following SC administration was characterised by first order absorption with a model estimated delay of 3 hours. No covariates were identified.

The bioavailability of natalizumab after subcutaneous administration was 84% as estimated using the updated population pharmacokinetic analysis. After SC administration of 300 mg natalizumab, peak values (C_{max}) were reached by approximately 1 week (t_{max}: 5.8 days, range from 2 to 7.9 days).

The mean C_{max} for RRMS participants was 35.44 µg/mL (range 22.0 to 47.8 µg/mL) being 33% of the peak values achieved following IV administration.

Multiple subcutaneous doses of 300 mg administered every 4 weeks resulted in comparable C_{trough} to 300 mg administered intravenously every 4 weeks. The predicted time to steady-state was approximately 24 weeks. In both intravenous and subcutaneous administration of natalizumab (every 4 weeks) C_{trough} values resulted in comparable α4β1 integrin binding.

Distribution

Both intravenous and subcutaneous routes of administration shared the same disposition PK parameters (CL, V_{ss} and $t_{½}$) and same sets of covariates as described in the updated population pharmacokinetic analysis.

Median steady-state volume of distribution was 5.58 L (5.27 to 5.92 L, 95% confidence interval).

Elimination

Population median estimate for linear clearance was 6.21 mL/h (5.60 to 6.70 mL/h, 95% confidence interval) and the estimated median half-life was 26.8 days. The 95th percentile interval of the terminal half-life is from 11.6 to 46.2 days.

The population analysis of 1 286 patients explored the effects of selected covariates including body weight, age, gender, presence of anti-natalizumab antibodies and formulation on pharmacokinetics. Only body weight, the presence of anti-natalizumab antibodies and the formulation used in phase 2 studies were found to influence natalizumab disposition. Natalizumab clearance increased with body weight in a less than proportional manner, such that a +/- 43% change in body weight resulted in only a -38% to 36% change in clearance. The presence of persistent anti-natalizumab antibodies increased natalizumab clearance approximately 2.54-fold, consistent with reduced serum natalizumab concentrations observed in persistently antibody-positive patients.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity and genotoxicity.

Consistent with the pharmacological activity of natalizumab, altered trafficking of lymphocytes was seen as white blood cell increases as well as increased spleen weights in most *in vivo* studies. These changes were reversible and did not appear to have any adverse toxicological consequences.

In studies conducted in mice, growth and metastasis of melanoma and lymphoblastic leukaemia tumour cells was not increased by the administration of natalizumab.

No clastogenic or mutagenic effects of natalizumab were observed in the Ames or human chromosomal aberration assays. Natalizumab showed no effects on *in vitro* assays of α4-integrin-positive tumour line proliferation or cytotoxicity.

Reductions in female guinea pig fertility were observed in one study at doses in excess of the human dose; natalizumab did not affect male fertility.

The effect of natalizumab on reproduction was evaluated in 5 studies, 3 in guinea pigs and 2 in *cynomolgus* monkeys. These studies showed no evidence of teratogenic effects or effects on growth of offspring. In one study in guinea pigs, a small reduction in pup survival was noted. In a study in monkeys, the number of abortions was doubled in the natalizumab 30 mg/kg treatment groups versus matching control groups. This was the result of a high incidence of abortions in treated groups in the first cohort that was not observed in the second cohort. No effects on abortion rates were noted in any other study. A study in pregnant *cynomolgus* monkeys demonstrated natalizumab-related changes in the foetus that included mild anaemia, reduced platelet counts, increased spleen weights and reduced liver and thymus weights. These changes were associated with increased splenic extramedullary haematopoiesis, thymic atrophy and decreased hepatic haematopoiesis. Platelet counts were also reduced in offspring born to mothers treated with natalizumab until parturition, however there was no evidence of anaemia in these offspring. All changes were observed at doses in excess of the human dose and were reversed upon clearance of natalizumab.

In *cynomolgus* monkeys treated with natalizumab until parturition, low levels of natalizumab were detected in the breast milk of some animals.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sodium phosphate, monobasic, monohydrate
Sodium phosphate, dibasic, heptahydrate
Sodium chloride
Polysorbate 80 (E 433)
Water for injections

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

6.3 Shelf life

3 years

6.4 Special precautions for storage

Store in a refrigerator (2 °C - 8 °C).
Do not freeze.
Keep the syringe in the outer carton in order to protect from light.

The pre-filled syringes can be kept at room temperature (up to 30 °C) for a maximum of a combined period of up to 24 hours, including the time to allow warming to room temperature for administration. The syringes can be returned to the refrigerator and used before the expiry date stated on the label and carton. Date and time of removal of the pack from the refrigerator must be recorded on the carton. Discard the syringes if left out of the refrigerator for more than 24 hours. Do not use external heat sources such as hot water to warm the pre-filled syringes.

6.5 Nature and contents of container

Each PFS consists of a pre-filled syringe made of glass (Type 1A) with a rubber stopper and thermoplastic rigid needle shield, containing 1 mL of solution. A 27 gauge needle is pre-affixed to the syringe. Each PFS has a needle guard system that will automatically cover the exposed needle when the plunger is fully depressed.

Pack sizes:
Pack containing two 150 mg pre-filled syringes, for one dose.
Multipack containing six 150 mg pre-filled syringes (3 packs of 2), for three doses.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MARKETING AUTHORISATION HOLDER

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

8. MARKETING AUTHORISATION NUMBERS

EU/1/06/346/002
EU/1/06/346/003

9. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

Date of first authorisation: 27th June 2006
Date of latest renewal: 18th April 2016

10. DATE OF REVISION OF THE TEXT

Detailed information on this medicinal product is available on the website of the European Medicines Agency https://www.ema.europa.eu.

ANNEX II

A. MANUFACTURERS OF THE BIOLOGICAL ACTIVE SUBSTANCE AND MANUFACTURER RESPONSIBLE FOR BATCH RELEASE

B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE

C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION

D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT

A. MANUFACTURERS OF THE BIOLOGICAL ACTIVE SUBSTANCE AND MANUFACTURER RESPONSIBLE FOR BATCH RELEASE

Name and address of the manufacturers of the biological active substance

Biogen MA Inc.
5000 Davis Drive
Morrisville
North Carolina 27560
USA

FUJIFILM Diosynth Biotechnologies Denmark ApS
Biotek Allé 1
DK-3400 Hillerød
Denmark

Biogen International GmbH
Attisholzstrasse 11
4542 Luterbach
Switzerland

Name and address of the manufacturer responsible for batch release

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE

Medicinal product subject to restricted medical prescription (See Annex I: Summary of Product Characteristics, section 4.2).

C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION

- **Periodic safety update reports (PSURs)**

The requirements for submission of PSURs for this medicinal product are set out in the list of Union reference dates (EURD list) provided for under Article 107c(7) of Directive 2001/83/EC and any subsequent updates published on the European medicines web-portal.

D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT

- **Risk management plan (RMP)**

The marketing authorisation holder (MAH) shall perform the required pharmacovigilance activities and interventions detailed in the agreed RMP presented in Module 1.8.2 of the marketing authorisation and any agreed subsequent updates of the RMP.

An updated RMP should be submitted:

- At the request of the European Medicines Agency;

- Whenever the risk management system is modified, especially as the result of new information being received that may lead to a significant change to the benefit/risk profile or as the result of an important (pharmacovigilance or risk minimisation) milestone being reached.

- **Additional risk minimisation measures**

Based on how patients treated with Tysabri are currently monitored at national level, the MAH shall discuss and agree with the National Competent Authorities measures to enhance further this monitoring (e.g. registries, post-marketing surveillance studies) as appropriate. The MAH shall implement agreed measures for monitoring within a time frame agreed with the National Competent Authorities.

The educational programme is aimed at educating healthcare professionals (HCPs) and patients/carers of the potential and risk factors for the development of PML, its diagnosis and treatment, and the identification and management of possible sequelae.

The MAH shall ensure that in each Member State where Tysabri is marketed, all healthcare professionals and patients/carers who are expected to prescribe/use Tysabri have access to/are provided with the educational materials listed below. Prior to implementation, the MAH must agree on the content and format of the educational materials, including communication media, distribution modalities, and any other aspects of the programme, with the National Competent Authority.

- Educational materials for HCPs:
  - Healthcare Professional Guide
  - For HCPs administering Tysabri SC Outside a Clinical Setting (OCS): Pre-Administration Checklist
- Patient information pack:
  - Patient Card
  - Treatment initiation and treatment continuation forms
  - Treatment discontinuation form
  - For patients and caregivers administering Tysabri SC: Pre-Administration Checklist

These educational materials shall contain the following key elements:

Healthcare Professional Guide:

- Background information on the increased risk of atypical/opportunistic infections, in particular PML, which may occur with Tysabri therapy, including a brief description of the **epidemiology, aetiology, and pathology** pertaining to the development of PML in Tysabri-treated patients.
- Information relating to the **identification of risk factors** for Tysabri-associated PML, including details of the PML risk estimates algorithm summarising PML risk by risk factor (anti-John Cunningham virus [JCV] antibody status, prior IS use, and duration of treatment [by year of treatment]), and stratification of this risk by index value when applicable.

- **Information on extending the dosing interval for PML risk mitigation**, including a reminder of the approved dosing schedule. The decrease in PML risk is based on data from the IV route of administration. No clinical data are available on either the safety or efficacy of dosing every 6 weeks with the SC route of administration.

- Inclusion of **monitoring guidance** for MRI and anti-JCV antibody based on PML risk, including recommended timing and interpretation of results.

- Information regarding the **diagnosis of PML**, including principles, clinical assessment (including MRI and laboratory testing), and differentiation between PML and MS.

- **Management** recommendations in the event of cases of suspected PML, including considerations on the effectiveness of PLEX treatment and the management of associated IRIS.

- Detail on the **prognosis** on PML, including information on improved outcomes observed in asymptomatic PML cases.

- A reminder that irrespective of the presence or absence of PML risk factors, heightened clinical vigilance for PML should be maintained in all patients treated with Tysabri and for 6 months following **discontinuation of therapy**.

- A statement that all data available to characterise PML risk are from the IV route of administration. Considering the similar PD profiles, the same PML risk and relevant risk factors are assumed for different routes of administration.

- A reminder on the need to discuss the benefit-risk profile of Tysabri treatment with the patient, and the requirement to provide the patient information pack.

- A reminder that it is the responsibility of the treating specialised physician to determine the patient's suitability for Tysabri SC administration OCS at regular intervals, and to ensure appropriate monitoring for PML (including risk factors and MRI screening).

- A statement that the administration of Tysabri SC OCS does not replace the need for regular contact with and clinical monitoring by the patient's treating specialised physician.

Pre-Administration Checklist:

- The Pre-Administration Checklist is designed to aid the administering HCPs and administering patients/caregivers in identifying risk factors for and early signs and symptoms of PML.

- The Pre-Administration Checklist is to be used by both, HCPs administering Tysabri SC OCS and by patients and caregivers administering Tysabri SC, and needs to be reviewed prior to each Tysabri SC administration.

- Guidance based on patient/caregiver checklist responses for escalation to the supervising specialised physician, whose responsibility it remains to determine next steps regarding the appropriateness and timing of Tysabri administration, if signs, symptoms, or new risk factors for PML are suspected.

- A statement that the checklist is not intended to be a substitute for consultation with the patient's treating specialised physician.

Patient Card:

- Reminder to patients to show the card to any doctor and/or caregiver involved with their treatment, and to keep the card with them for 6 months after the last dose of Tysabri treatment.
- Reminder to patients to read the package leaflet carefully before starting Tysabri, and not to start Tysabri if there is a serious problem with their immune system.
- Reminder to patients not to take any other long-term medicines for MS while receiving Tysabri.
- A description of PML, potential symptoms and management of PML.
- A reminder of where to report side effects.
- Details of the patient, treating doctor and date Tysabri was started.
- Reminder to patients self-administering and caregivers administering Tysabri SC to review the Pre-Administration Checklist prior to each administration of Tysabri SC. If any symptoms of PML are noted, Tysabri SC may not be administered, and the prescriber needs to be informed immediately.

Treatment initiation and treatment continuation forms:

- Information on PML and IRIS, including the risk of developing PML during Tysabri treatment stratified by prior treatment with immunosuppressants and JCV infection.
- Confirmation that the doctor has discussed the risks of PML and the risk of IRIS if treatment is discontinued following suspicion of PML, and confirmation of patient understanding of the risks of PML and that they have received a copy of the treatment initiation form and a Patient Card.
- Patient details and prescriber name.
- The treatment continuation form should contain the elements of the treatment initiation form and, in addition, a statement that the risks of PML increase with duration of treatment and that treatment beyond 24 months carries additional risk.

Treatment discontinuation form

- Information for the patient that PML has been reported up to 6 months after stopping Tysabri, and to therefore keep the Patient Card with them after treatment discontinuation.
- Reminder of PML symptoms, and when MRI imaging may be warranted.
- Reporting of side effects.

ANNEX III

LABELLING AND PACKAGE LEAFLET

A. LABELLING

PARTICULARS TO APPEAR ON THE OUTER PACKAGING

OUTER CARTON

1. NAME OF THE MEDICINAL PRODUCT

TYSABRI 300 mg concentrate for solution for infusion
natalizumab

2. STATEMENT OF ACTIVE SUBSTANCE

Each 15 mL vial of concentrate contains 300 mg natalizumab (20 mg per mL). When diluted the solution for infusion contains approximately 2.6 mg/mL of natalizumab.

3. LIST OF EXCIPIENTS

sodium phosphate, monobasic, monohydrate; sodium phosphate, dibasic, heptahydrate; sodium chloride; polysorbate 80 (E 433) and water for injections.

Read the package leaflet before use.

4. PHARMACEUTICAL FORM AND CONTENTS

concentrate for solution for infusion
1 x 15 mL vial

5. METHOD AND ROUTE OF ADMINISTRATION

Intravenous use.
Dilute before infusion.
After dilution, do not shake.

Read the package leaflet before use.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN

7. OTHER SPECIAL WARNING(S), IF NECESSARY

8. EXPIRY DATE

EXP

9. SPECIAL STORAGE CONDITIONS

Store in a refrigerator. Do not freeze. Keep the vial in the outer carton in order to protect from light.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

12. MARKETING AUTHORISATION NUMBER

EU/1/06/346/001

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Justification for not including Braille accepted.

17. UNIQUE IDENTIFIER – 2D BARCODE

2D barcode carrying the unique identifier included.

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA

PC
SN
NN

MINIMUM PARTICULARS TO APPEAR ON SMALL IMMEDIATE PACKAGING UNITS

VIAL LABEL

1. NAME OF THE MEDICINAL PRODUCT AND ROUTE OF ADMINISTRATION

TYSABRI 300 mg concentrate for solution for infusion
natalizumab
IV

2. METHOD OF ADMINISTRATION

Dilute before infusion. After dilution, do not shake.

Read the package leaflet before use.

3. EXPIRY DATE

EXP

4. BATCH NUMBER

Lot

5. CONTENTS BY WEIGHT, BY VOLUME OR BY UNIT

15 mL

6. OTHER

PARTICULARS TO APPEAR ON THE OUTER PACKAGING

CARTON

1. NAME OF THE MEDICINAL PRODUCT

Tysabri 150 mg solution for injection in pre-filled syringe
natalizumab

2. STATEMENT OF ACTIVE SUBSTANCE

Each pre-filled syringe contains 150 mg natalizumab in 1 mL solution

3. LIST OF EXCIPIENTS

sodium phosphate, monobasic, monohydrate; sodium phosphate, dibasic, heptahydrate; sodium chloride; polysorbate 80 (E 433) and water for injections.

4. PHARMACEUTICAL FORM AND CONTENTS

Solution for injection
2 pre-filled syringes

5. METHOD AND ROUTE OF ADMINISTRATION

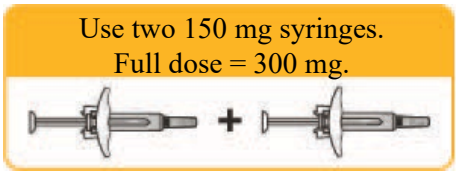
Read the package leaflet before use.
Subcutaneous use.
For single use only.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN

Keep out of the sight and reach of children.

7. OTHER SPECIAL WARNING, IF NECESSARY

Use two 150 mg syringes.
Full dose = 300 mg.



8. EXPIRY DATE

EXP

9. SPECIAL STORAGE CONDITIONS

Store in a refrigerator. Do not freeze.
Store the pre-filled syringe in the original package in order to protect from light

Syringes can be left at room temperature (up to 30 °C) for up to 24 hours in total.

Record total time outside of the refrigerator.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

12. MARKETING AUTHORISATION NUMBER

EU/1/06/346/002

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Tysabri 150 mg

17. UNIQUE IDENTIFIER – 2D BARCODE

2D barcode carrying the unique identifier included.

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA

PC
SN
NN

PARTICULARS TO APPEAR ON THE OUTER PACKAGING

MULTIPACK OUTER CARTON

Pre-filled syringe 150 mg Multipack (with bluebox)

1. NAME OF THE MEDICINAL PRODUCT

Tysabri 150 mg solution for injection in pre-filled syringe
natalizumab

2. STATEMENT OF ACTIVE SUBSTANCES

Each pre-filled syringe contains 150 mg natalizumab in 1 mL solution

3. LIST OF EXCIPIENTS

sodium phosphate, monobasic, monohydrate; sodium phosphate, dibasic, heptahydrate; sodium chloride; polysorbate 80 (E 433) and water for injections.

4. PHARMACEUTICAL FORM AND CONTENTS

Solution for injection

Multipack: 6 (3 packs of 2) pre-filled syringes of 150 mg.

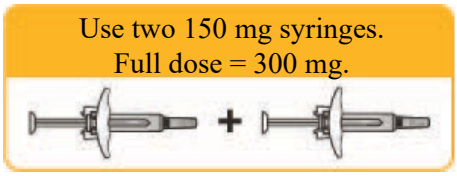
5. METHOD AND ROUTES OF ADMINISTRATION

Read the package leaflet before use.
Subcutaneous use.
For single use only.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN

Keep out of the sight and reach of children.

7. OTHER SPECIAL WARNING, IF NECESSARY



8. EXPIRY DATE

EXP

9. SPECIAL STORAGE CONDITIONS

Store in a refrigerator. Do not freeze.
Store the pre-filled syringe in the original package in order to protect from light

Syringes can be left at room temperature (up to 30 °C) for up to 24 hours in total.

Record total time outside of the refrigerator.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

12. MARKETING AUTHORISATION NUMBER

EU/1/06/346/003

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Tysabri 150 mg

17. UNIQUE IDENTIFIER – 2D BARCODE

2D barcode carrying the unique identifier included.

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA

PC
SN
NN

PARTICULARS TO APPEAR ON THE OUTER PACKAGING

MULTIPACK INNER CARTON

Pre-filled syringe 150 mg Multipack (without bluebox)

1. NAME OF THE MEDICINAL PRODUCT

Tysabri 150 mg solution for injection in pre-filled syringe
natalizumab

2. STATEMENT OF ACTIVE SUBSTANCES

Each pre-filled syringe contains 150 mg natalizumab in 1 mL solution

3. LIST OF EXCIPIENTS

sodium phosphate, monobasic, monohydrate; sodium phosphate, dibasic, heptahydrate; sodium chloride; polysorbate 80 (E 433) and water for injections.

4. PHARMACEUTICAL FORM AND CONTENTS

Solution for injection
2 pre-filled syringes. Component of a multipack, cannot be sold separately.

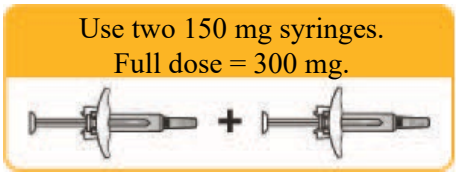
5. METHOD AND ROUTES OF ADMINISTRATION

Read the package leaflet before use.
Subcutaneous use.
For single use only.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN

Keep out of the sight and reach of children.

7. OTHER SPECIAL WARNING, IF NECESSARY



8. EXPIRY DATE

EXP

9. SPECIAL STORAGE CONDITIONS

Store in a refrigerator. Do not freeze.
Store the pre-filled syringe in the original package in order to protect from light

Syringes can be left at room temperature (up to 30 °C) for up to 24 hours in total.

Record total time outside of the refrigerator.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

12. MARKETING AUTHORISATION NUMBER

EU/1/06/346/003

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Tysabri 150 mg

17. UNIQUE IDENTIFIER – 2D BARCODE

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA

PARTICULARS TO APPEAR ON THE OUTER PACKAGING

SYRINGE TRAY

1. OTHER

Use two 150 mg syringes
Full dose = 300 mg

Text to be contained on peel-off portion:
2 x Tysabri 150 mg SC

Lot

EXP

MINIMUM PARTICULARS TO APPEAR ON SMALL IMMEDIATE PACKAGING UNITS
PRE-FILLED SYRINGE LABEL

1. NAME OF THE MEDICINAL PRODUCT AND ROUTE OF ADMINISTRATION

Tysabri 150 mg injection
natalizumab
SC

2. METHOD OF ADMINISTRATION

3. EXPIRY DATE

EXP

4. BATCH NUMBER

Lot

5. CONTENTS BY WEIGHT, BY VOLUME OR BY UNIT

1 mL

6. OTHER

B. PACKAGE LEAFLET

Package leaflet: Information for the patient

Tysabri 300 mg concentrate for solution for infusion
natalizumab

Read all of this leaflet carefully before you start using this medicine because it contains important information for you.

In addition to this leaflet you will be given a Patient Card. This contains important safety information that you need to know before and during treatment with Tysabri.

- Keep this leaflet and the Patient Card. You may need to read them again. Keep the leaflet and Patient Card with you during treatment and for six months after the last dose of this medicine , as side effects may occur even after you have stopped treatment.
- If you have any further questions, ask your doctor.
- If you get any side effects talk to your doctor. This includes any possible side effects not listed in this leaflet. See section 4.

What is in this leaflet:

1. **What Tysabri is and what it is used for**
2. **What you need to know before you receive Tysabri**
3. **How Tysabri is given**
4. **Possible side effects**
5. **How to store Tysabri**
6. **Contents of the pack and other information**

1. What Tysabri is and what it is used for

Tysabri is used to treat multiple sclerosis (MS) in adults. It contains the active substance natalizumab. This is called a *monoclonal antibody*.

MS causes inflammation in the brain that damages the nerve cells. This inflammation happens when white blood cells get into the brain and spinal cord. This medicine stops the white blood cells getting through to the brain. This reduces nerve damage caused by MS.

Symptoms of multiple sclerosis
The symptoms of MS vary from patient to patient, and you may experience some or none of them.

They may include: walking problems, numbness in the face, arms or legs; problems with vision; tiredness; feeling off-balance or light headed; bladder and bowel problems; difficulty in thinking and concentrating; depression; acute or chronic pain; sexual problems; stiffness and muscle spasms. When the symptoms flare up, it is called a *relapse* (also known as an exacerbation or an attack). When a relapse occurs, you may notice the symptoms suddenly, within a few hours, or slowly progressing over several days. Your symptoms will then usually improve gradually (this is called a remission).

How Tysabri can help
In trials, this medicine approximately halved the build-up of disability caused by MS, and decreased the number of MS attacks by about two-thirds. While you are treated with this medicine you might not notice any improvement, but it may still be working to prevent your MS becoming worse.

2. What you need to know before you receive Tysabri

Before you start treatment with this medicine, it is important that you and your doctor have discussed the benefits you could expect to receive from this treatment and the risks that are associated with it.

You must not be given Tysabri

- If you are **allergic** to natalizumab or any of the other ingredients of this medicine (listed in section 6).
- If you have been **diagnosed with PML** (*progressive multifocal leukoencephalopathy*). PML is an uncommon infection of the brain.
- If your **immune system** has a serious problem. This may be due to disease (such as HIV), or to a medicine you are taking, or have taken in the past (see below).
- If you are taking **medicines that affect your immune system,** including certain other medicines used to treat MS. These medicines cannot be used with Tysabri.
- If you **have cancer** (unless it is a type of skin cancer called *basal cell carcinoma*).

Warnings and precautions

You need to discuss with your doctor whether Tysabri is the most suitable treatment for you. Do this before you start taking Tysabri, and when you have been receiving Tysabri for more than two years.

Possible brain infection (PML)

Some people receiving this medicine (fewer than 1 in 100) have had an uncommon brain infection called PML (*progressive multifocal leukoencephalopathy*). PML can lead to severe disability or death.

- Before starting treatment, **all patients will have blood tests** arranged by the doctor for JC virus infection. JC virus is a common virus that does not normally make you ill. However, PML is linked to an increase of JC virus in the brain. The reason for this increase in some patients treated with Tysabri is not clear. Before and during treatment, your doctor will test your blood to check if you have antibodies to the JC virus, which are a sign that you have been infected by the JC virus.
- Your doctor will arrange a **Magnetic Resonance Imaging (MRI) scan**, which will be repeated during treatment to rule out PML.
- **The symptoms of PML** may be similar to an MS relapse (see section 4, *Possible side effects*). You can also get PML up to 6 months after stopping Tysabri treatment.

  Tell your doctor as soon as possible if you notice your MS getting worse, if you notice any new symptoms while you are on Tysabri treatment or for up to 6 months afterwards.
- **Tell your partner or caregivers** about what to look out for (see also section 4, *Possible side effects*). Some symptoms might be difficult to spot by yourself, such as changes in mood or behaviour, confusion, speech and communication difficulties. If you get any of these, **you may need further tests**. Keep looking out for symptoms in the 6 months after stopping Tysabri.
- Keep the Patient Card you have been given by your doctor. It includes this information. Show it to your partner or caregivers.

Three things can increase your risk of PML with Tysabri. If you have two or more of these risk factors, the risk is increased further:

- **If you have antibodies to the JC virus** in your blood. These are a sign that the virus is in your body. You will be tested before and during Tysabri treatment.
- **If you are treated for a long time** with Tysabri, especially if it is more than two years.
- **If you have taken a medicine called an *immunosuppressant***, that reduces the activity of your immune system.

Another condition, called JCV GCN (*JC virus granule cell neuronopathy*), is also caused by JC virus and has occurred in some patients receiving Tysabri. The symptoms of JCV GNC are similar to PML.

For those with a lower risk of PML, your doctor may repeat the test regularly to check that:

- You still do not have antibodies to the JC virus in your blood.
- If you have been treated for more than 2 years, you still have a lower level of JC virus antibodies in your blood.

If someone gets PML
PML can be treated, and Tysabri treatment will be stopped. However, some people get a reaction as Tysabri is removed from the body. This reaction (known as IRIS or immune reconstitution inflammatory syndrome) may lead to your condition getting worse, including worsening of brain function.

Look out for other infections
Some infections other than PML may also be serious and can be due to viruses, bacteria, and other causes.

Tell a doctor or nurse immediately if you think you have an infection (see also section 4, *Possible side effects*).

Changes in blood platelets
Natalizumab may reduce platelets in the blood which are responsible for clotting. This may result in a condition called thrombocytopenia (see section 4) in which your blood may not clot quickly enough to stop bleeding. This can lead to bruising as well as other more serious problems such as excessive bleeding. You should talk to your doctor immediately if you have unexplained bruising, red or purple spots on the skin (called petechiae), bleeding from skin cuts that does not stop or oozes, prolonged bleeding from the gums or nose, blood in urine or stools, or bleeding in the whites of your eyes.

Children and adolescents
Do not give this medicine to children or adolescents under the age of 18 years.

Other medicines and Tysabri
Tell your doctor if you are taking, have recently taken or might take any other medicines.

- You **must not** be given this medicine if you are now being treated with medicines that affect your **immune system**, including certain other medicines to treat your MS.
- You might not be able to use this medicine if you have **previously** had any medicines that affect your immune system.

Pregnancy and breast-feeding

- **Do not use this medicine if you are pregnant,** unless you have discussed this with your doctor. Be sure to tell your doctor immediately if you get pregnant, think you may be pregnant, or if you are planning to become pregnant.

- **Do not breast-feed whilst using Tysabri.** Your doctor will help you decide whether you should choose to stop breast-feeding or stop using the medicine.

If you are pregnant or breast-feeding, think you may be pregnant or are planning to have a baby, ask your doctor for advice before taking this medicine. The risk to the baby and benefit to the mother will be taken into consideration by your doctor.

Driving and using machines
Dizziness is a very common side effect. If you are affected, do not drive or use machines.

Tysabri contains polysorbate 80 (E 433)
This medicine contains 3 mg of polysorbate 80 in each vial. Polysorbates may cause allergic reactions. Tell your doctor if you have any known allergies.

Tysabri contains sodium
Each vial of this medicine contains 2.3 mmol (or 52 mg) of sodium. After dilution for use, this medicinal product contains 17.7 mmol (or 406 mg) sodium per dose. This should be considered if you are on a controlled sodium diet.

3. How Tysabri is given

Tysabri IV infusion will be given to you by a doctor experienced in the treatment of MS. Your doctor may switch you directly from another medicine for MS to Tysabri if there are no problems caused by your previous treatment.

- Your doctor will order **blood tests** for antibodies to the JC virus and other possible problems.

- Your doctor will arrange an **MRI scan**, which will be repeated during treatment.

- **To switch from some MS medicines,** your doctor may advise you to wait for a certain time to ensure that most of the previous medicine has left your body.

- For adults the recommended dose is 300 mg, given once every 4 weeks.

- Tysabri must be diluted before it is given to you. It is given as a drip into a vein (by intravenous infusion), usually in your arm. This takes about 1 hour.

- Information for medical or healthcare professionals on how to prepare and administer the medicine is provided at the end of this leaflet.

If you stop using Tysabri
Regular dosing with Tysabri is important, especially in the first few months of treatment. It is important to continue with your medicine for as long as you and your doctor decide that it is helping you. Patients who received one or two doses of Tysabri, and then had a gap in treatment of three months or more, were more likely to have an allergic reaction when restarting treatment.

Checking for allergic reactions
A few patients have had an allergic reaction to this medicine. Your doctor may check for allergic reactions during the infusion and for 1 hour afterwards. See also section 4, *Possible side effects*.

If you miss your dose of Tysabri
If you miss your usual dose of Tysabri, arrange with your doctor to receive it as soon as you can. You can then continue to receive your dose of Tysabri every 4 weeks.

Will Tysabri always work?
In a few patients receiving Tysabri, the body's natural defences may stop the medicine from working properly over time, as the body develops antibodies to the medicine. Your doctor can decide whether this medicine is not working properly for you from blood tests and will stop the treatment, if necessary.

If you have any further questions on Tysabri, ask your doctor. Always use this medicine exactly as described in this leaflet or as your doctor has told you. Check with your doctor if you are not sure.

4. Possible side effects

Like all medicines, this medicine can cause side effects, although not everybody gets them.

Speak to your doctor or nurse immediately if you notice any of the following.

Signs of a brain infection
- Changes in personality and behaviour such as confusion, delirium or loss of consciousness,
- Seizures (fits)
- Headache
- Nausea / vomiting
- Stiff neck
- Extreme sensitivity to bright light
- Fever
- Rash (anywhere on the body)

These symptoms may be caused by an infection of the brain (*encephalitis or PML*) or its covering layer (*meningitis*).

Signs of other serious infections
- An unexplained fever
- Severe diarrhoea
- Shortness of breath
- Prolonged dizziness
- Headache
- Weight loss
- Listlessness
- Impaired vision
- Pain or redness of the eye(s)

Signs of an allergic reaction
- Itchy rash (*hives*)
- Swelling of your face, lips or tongue
- Difficulty breathing
- Chest pain or discomfort
- Increase or decrease in your blood pressure (your doctor or nurse will notice this if they are monitoring your blood pressure)

These are most likely during or shortly after the infusion.

Signs of a possible liver problem
- Yellowing of your skin or the whites of your eyes
- Unusual darkening of the urine
- Abnormal liver function test

Speak to a doctor or nurse immediately if you get any of the side effects listed above, or if you think you have an infection. **Show your Patient Card** and this package leaflet to any doctor or nurse who treats you, not only to your neurologist.

Other side effects
Very common (may affect more than 1 in 10 people)
- Urinary tract infection
- Sore throat and runny or blocked up nose
- Headache
- Dizziness
- Feeling sick (*nausea*)
- Joint pain
- Tiredness
- Dizziness, feeling sick (*nausea*), itching and chills during or shortly after infusion

Common (may affect up to 1 in 10 people)
- Anaemia (decrease in your red blood cells which can make your skin pale and can make you feel breathless or lacking energy)
- Allergy (*hypersensitivity*)
- Shivering
- Itchy rash (*hives*)
- Being sick (*vomiting*)
- Fever
- Difficulty breathing *(dyspnoea)*
- Reddening of the face or body (*flushing*)
- Herpes infections
- Discomfort around the place you have had your infusion. You could experience bruising, redness, pain, itching or swelling

Uncommon (may affect up to 1 in 100 people)
- Severe allergy (*anaphylactic reaction*)
- Progressive multifocal leukoencephalopathy (PML)
- Inflammatory disorder after discontinuation of the medicinal product
- Facial swelling
- An increase in the number of white blood cells (*eosinophilia*)
- Reduction in blood platelets
- Easy bruising (purpura)

Rare (may affect up to 1 in 1 000 people)
- Herpes infection in the eye
- Severe anaemia (decrease in your red blood cells which can make your skin pale and can make you feel breathless or lacking energy).
- Severe swelling under the skin
- High levels of bilirubin in the blood (*hyperbilirubinaemia*) which may cause symptoms such as yellowing of your eyes or skin, fever and tiredness

Not known (frequency cannot be estimated from the available data)
- Unusual infections (so-called "*opportunistic infections*")
- Damage to your liver

Speak to your doctor as soon as possible if you think you have an infection.
You will also find this information in the Patient Card you have been given by your doctor.

Reporting of side effects
If you get any side effects, talk to your doctor. This includes any possible side effects not listed in this leaflet. You can also report side effects directly via the national reporting system listed in Appendix V.
By reporting side effects you can help provide more information on the safety of this medicine.

5. How to store Tysabri

Keep this medicine out of the sight and reach of children.

Do not use this medicine after the expiry date stated on the label and carton. The expiry date refers to the last day of that month.

Unopened vial:
Store in a refrigerator .
Do not freeze.
Keep the vial in the outer carton in order to protect from light.

Diluted solution:
After dilution, immediate use is recommended. If not used immediately, the diluted solution must be stored at 2 °C to 8 °C and infused within 24 hours of dilution.

Do not use this medicine if you notice particles in the liquid and/or the liquid in the vial is discoloured.

6. Contents of the pack and other information

What Tysabri contains
The active substance is natalizumab. Each 15 mL vial of concentrate contains 300 mg natalizumab (20 mg per mL). When diluted, the solution for infusion contains approximately 2.6 mg per mL of natalizumab.

The other ingredients are:
Sodium phosphate, monobasic, monohydrate,
Sodium phosphate, dibasic, heptahydrate,
Sodium chloride (see section 2 ‘Tysabri contains sodium’),
Polysorbate 80 (E 433)
Water for injections

What Tysabri looks like and contents of the pack
Tysabri is a clear, colourless to slightly cloudy liquid.
Each carton contains one glass vial.

Marketing Authorisation Holder and Manufacturer
Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

For any further information about this medicine, please contact the local representative of the Marketing Authorisation Holder.

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This leaflet was last revised in .

Other sources of information

Detailed information on this medicine is available on the European Medicines Agency web site: https://www.ema.europa.eu.

The following information is intended for healthcare professionals only:

1. Inspect the Tysabri vial for particles prior to dilution and administration. If particles are observed and/or the liquid in the vial is not colourless, clear to slightly opalescent, the vial must not be used.

2. Use aseptic technique when preparing the medicine. Remove flip-top from the vial. Insert the syringe needle into the vial through the centre of the rubber stopper and remove 15 mL concentrate for solution for infusion.

3. Add the 15 mL concentrate for solution for infusion to 100 mL sodium chloride 9 mg/mL (0.9%) solution for injection. Gently invert the solution to mix completely. Do not shake.

4. Tysabri must not be mixed with other medicinal products or diluents.

5. Visually inspect the diluted medicinal product for particles or discolouration prior to administration. Do not use if it is discoloured or if foreign particles are seen.

6. The diluted medicinal product is to be used as soon as possible and within 24 hours of dilution. If the diluted medicinal product is stored at 2 °C to 8 °C (do not freeze), allow the solution to warm to room temperature prior to infusion.

7. The diluted solution is to be infused intravenously over 1 hour at a rate of approximately 2 mL per minute.

8. After the infusion is complete, flush the intravenous line with sodium chloride 9 mg/mL (0.9%) solution for injection.

9. Each vial is for single–use only.

10. In order to improve traceability of biological medicinal products, the product name (Tysabri) and batch number of the administered product should be clearly recorded.

11. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Package leaflet: Information for the patient

Tysabri 150 mg solution for injection in pre-filled syringe
natalizumab

Read all of this leaflet carefully before you start using this medicine because it contains important information for you.

In addition to this leaflet you will be given a Patient Card and, in case of self-administration or administration by a caregiver, a Pre-Administration Checklist. These contain important safety information that you need to know before and during treatment with Tysabri.

- Keep this leaflet and the Patient Card. You may need to read them again. Keep the leaflet and Patient Card with you during treatment and for six months after the last dose of this medicine, as side effects may occur even after you have stopped treatment. If you or your caregiver administer the treatment, review the Pre-Administration Checklist before each dose.
- If you have any further questions, ask your doctor.
- This medicine has been prescribed for you only. Do not pass it on to others. It may harm them, even if their signs of illness are the same as yours.
- If you get any side effects talk to your doctor. This includes any possible side effects not listed in this leaflet. See section 4.

What is in this leaflet:

1. **What Tysabri is and what it is used for**
2. **What you need to know before you receive Tysabri**
3. **How Tysabri is given**
4. **Possible side effects**
5. **How to store Tysabri**
6. **Contents of the pack and other information**

1. What Tysabri is and what it is used for

Tysabri is used to treat multiple sclerosis (MS) in adults. It contains the active substance natalizumab. This is called a *monoclonal antibody*.

MS causes inflammation in the brain that damages the nerve cells. This inflammation happens when white blood cells get into the brain and spinal cord. This medicine stops the white blood cells getting through to the brain. This reduces nerve damage caused by MS.

Symptoms of multiple sclerosis
The symptoms of MS vary from patient to patient, and you may experience some or none of them. They may include: walking problems; numbness in the face, arms or legs; problems with vision; tiredness; feeling off-balance or light headed; bladder and bowel problems; difficulty in thinking and concentrating; depression; acute or chronic pain; sexual problems; stiffness and muscle spasms.

When the symptoms flare up, it is called a *relapse* (also known as an exacerbation or an attack). When a relapse occurs, you may notice the symptoms suddenly, within a few hours, or slowly progressing over several days. Your symptoms will then usually improve gradually (this is called a *remission*).

How Tysabri can help
In trials**,** this medicine approximately halved the build-up of disability caused by MS, and decreased the number of MS attacks by about two-thirds. While you are treated with this medicine you might not notice any improvement, but it may still be working to prevent your MS becoming worse.

2. What you need to know before you receive Tysabri

Before you start treatment with this medicine, it is important that you and your doctor have discussed the benefits you could expect to receive from this treatment and the risks that are associated with it.

You must not be given Tysabri

- If you are **allergic** to natalizumab or any of the other ingredients of this medicine (listed in section 6).
- If you have been **diagnosed with PML** (*progressive multifocal leukoencephalopathy*). PML is an uncommon infection of the brain.
- If your **immune system** has a serious problem. This may be due to disease (such as HIV), or to a medicine you are taking, or have taken in the past (see below).
- If you are taking **medicines that affect your immune system**, including certain other medicines used to treat MS. These medicines cannot be used with Tysabri.
- If you **have cancer** (unless it is a type of skin cancer called *basal cell carcinoma*).

Warnings and precautions
You need to discuss with your doctor whether Tysabri is the most suitable treatment for you. Do this before you start taking this medicine, and when you have been receiving it for more than two years.

Keeping a record
In order to improve the traceability of this medicine, your doctor or pharmacist should record the name and the lot number of the product you have been given in your patient file. You may also wish to make a note of these details in case you are asked for this information in the future.

Possible brain infection (PML)
Some people receiving this medicine (fewer than 1 in 100) have had an uncommon brain infection called PML (*progressive multifocal leukoencephalopathy*). PML can lead to severe disability or death.

- Before starting treatment, **all patients will have blood tests** arranged by the doctor for JC virus infection. JC virus is a common virus that does not normally make you ill. However, PML is linked to an increase of JC virus in the brain. The reason for this increase in some patients treated with Tysabri is not clear. Before and during treatment, your doctor will test your blood to check if you have antibodies to the JC virus, which are a sign that you have been infected by the JC virus.
- Your doctor will arrange a **Magnetic Resonance Imaging (MRI) scan**, which will be repeated during treatment to rule out PML.
- **The symptoms of PML** may be similar to an MS relapse (see section 4, *Possible side effects*). You can also get PML up to 6 months after stopping Tysabri treatment.

  Tell your doctor as soon as possible if you notice your MS getting worse or if you notice any new symptoms, while you are on Tysabri treatment or for up to 6 months afterwards.
- **Tell your partner or caregivers** about what to look out for (see also section 4, *Possible side effects*). Some symptoms might be difficult to spot by yourself, such as changes in mood or behaviour, confusion, speech and communication difficulties. If you get any of these, **you may need further tests**. Keep looking out for symptoms in the 6 months after stopping Tysabri.

- Keep the Patient Card you have been given by your doctor. It includes this information. Show it to your partner or caregivers.

- If you or your caregiver administer the treatment, review the Pre-Administration Checklist **before each dose**.

Three things can increase your risk of PML with Tysabri. If you have two or more of these risk factors, the risk is increased further:

- **If you have antibodies to the JC virus** in your blood. These are a sign that the virus is in your body. You will be tested before and during Tysabri treatment.

- **If you are treated for a long time** with Tysabri, especially if it is more than two years.

- **If you have taken a medicine called an *immunosuppressant***, that reduces the activity of your immune system.

Another condition, called JCV GCN (*JC virus granule cell neuronopathy*), is also caused by JC virus and has occurred in some patients receiving this medicine. The symptoms of JCV GCN are similar to PML.

For those with a lower risk of PML, your doctor may repeat the test regularly to check that:

- You still do not have antibodies to the JC virus in your blood.

- If you have been treated for more than 2 years, you still have a lower level of JC virus antibodies in your blood.

If someone gets PML
PML can be treated, and Tysabri treatment will be stopped. However, some people **get a reaction** as Tysabri is removed from the body. This reaction (known as **IRIS**, or *immune reconstitution inflammatory syndrome*) may lead to your condition getting worse, including worsening of brain function.

Look out for other infections
Some infections other than PML may also be serious and can be due to viruses, bacteria, and other causes.

Tell a doctor or nurse immediately if you think you have an infection (see also section 4, *Possible side effects*).

Changes in blood platelets
Natalizumab may reduce platelets in the blood which are responsible for clotting. This may result in a condition called thrombocytopenia (see section 4) in which your blood may not clot quickly enough to stop bleeding. This can lead to bruising as well as other more serious problems such as excessive bleeding. You should talk to your doctor immediately if you have unexplained bruising, red or purple spots on the skin (called petechiae), bleeding from skin cuts that does not stop or oozes, prolonged bleeding from the gums or nose, blood in urine or stools, or bleeding in the whites of your eyes.

Children and adolescents
Do not give this medicine to children or adolescents under the age of 18 years.

Other medicines and Tysabri
Tell your doctor if you are taking, have recently taken or might take any other medicines.

- **You must not be given this medicine** if you are now being treated with medicines that affect your **immune system**, including certain other medicines to treat your MS.

- You might not be able to use this medicine if you have **previously** had any medicines that affect your immune system.

Pregnancy and breast-feeding

- **Do not use this medicine if you are pregnant**, unless you have discussed this with your doctor. Be sure to tell your doctor immediately if you get pregnant, think you may be pregnant, or if you are planning to become pregnant.

- **Do not breast-feed whilst using Tysabri**. Your doctor will help you decide whether you should stop breast-feeding, or stop using the medicine.

If you are pregnant or breast-feeding, think you may be pregnant or are planning to have a baby, ask your doctor for advice before taking this medicine. The risk to the baby and benefit to the mother will be taken into consideration by your doctor.

Driving and using machines
Dizziness is a very common side effect. If you are affected, do not drive or use machines.

Tysabri contains polysorbate 80 (E 433)
This medicine contains 0.4 mg of polysorbate 80 in each pre-filled syringe, which is equivalent to 0.8 mg per dose. Polysorbates may cause allergic reactions. Tell your doctor if you have any known allergies.

Tysabri contains sodium
This medicine contains less than 1 mmol sodium (23 mg) per 300 mg dose, so it is essentially 'sodium-free'.

3. How Tysabri is given

Tysabri injections will be prescribed to you by a doctor experienced in the treatment of MS. Your doctor may switch you directly from another medicine to Tysabri if there are no signs of problems caused by your previous treatment.

- Your doctor will order **blood tests** for antibodies to the JC virus and other possible problems.

- Your doctor will arrange an **MRI scan**, which will be repeated during treatment.

- **To switch from some MS medicines,** your doctor may advise you to wait for a certain time to ensure that most of the previous medicine has left your body.

- If your condition allows, your doctor may discuss with you the option of receiving injections outside a clinic (e.g. at home). These injections can be administered by a healthcare professional, by yourself or a caregiver, provided you meet certain criteria. **You will still need to attend the clinic or hospital for appointments including those for regular blood tests and MRI scans**.

- If your doctor decides that you are suitable for self-administration (or administration by your caregiver), a healthcare professional will supervise you for the administration of the first two doses (2 injections each).

- Your healthcare professional will give you or your caregiver detailed instructions and will show you how to prepare and inject the medicine before using the syringes the first time.

- If your doctor decides that you are suitable for administration by yourself or a caregiver, make sure you read the Patient Card to review the list of PML symptoms and you review the Pre-Administration Checklist **before each dose**. If any symptoms appear or worsen, do not administer the dose and contact your doctor immediately.

- For adults the recommended dose is 300 mg, given once every 4 weeks.

- Each dose is given as **two injections** under the skin, in your thigh, abdomen (at least 6 centimetres away from the belly button) or back of your arm (the latter only in case of injection by a healthcare professional or a caregiver). This takes up to 30 minutes.

- Information on how to prepare and inject the medicine is provided at the end of this leaflet.

If you stop using Tysabri
Regular dosing with this medicine is important, especially in the first few months of treatment. It is important to continue with your medicine for as long as you and your doctor decide that it is helping you. **Do not stop using your medicine without your doctor's advice**. Patients who received one or two doses of Tysabri, and then had a gap in treatment of 3 months or more, were more likely to have an allergic reaction when restarting treatment.

Checking for allergic reactions
A few patients have had an allergic reaction to this medicine. Your doctor may check for allergic reactions during the injections and for 1 hour afterwards. In case of self-administration or administration by a caregiver, if you experience an allergic reaction stop the injection and seek medical attention immediately. See also section 4, *Possible side effects*.

If you miss your dose of Tysabri
If you miss your usual dose of Tysabri, arrange with your doctor to receive it as soon as you can. You can then continue to receive your dose of Tysabri every 4 weeks.
Two syringes need to be administered to give the full dose. It is important that **both syringes** are administered and that you have them at the prescribed dosing schedule. If you or your caregiver are administering the injections and you have missed a dose or you have injected only one syringe, contact your doctor as soon as possible for advice.

Will Tysabri always work?
In a few patients receiving Tysabri, the body's natural defences may stop the medicine from working properly over time, as the body develops antibodies to the medicine. Your doctor can decide whether this medicine is not working properly for you from blood tests and will stop the treatment, if necessary.

If you have any further questions on Tysabri, ask your doctor. Always use this medicine exactly as described in this leaflet or as your doctor has told you. Check with your doctor if you are not sure.

Subcutaneous is abbreviated as SC on the syringe label.

4. Possible side effects

Like all medicines, this medicine can cause side effects, although not everybody gets them.

Speak to your doctor or nurse immediately if you notice any of the following.

Signs of a brain infection
- Changes in personality and behaviour such as confusion, delirium or loss of consciousness,
- Seizures (fits)
- Headache
- Nausea / vomiting
- Stiff neck

- Extreme sensitivity to bright light
- Fever
- Rash (anywhere on the body)

These symptoms may be caused by an infection of the brain (*encephalitis or PML*) or its covering layer (*meningitis*).

Signs of other serious infections
- An unexplained fever
- Severe diarrhoea
- Shortness of breath
- Prolonged dizziness
- Headache
- Weight loss
- Listlessness
- Impaired vision
- Pain or redness of the eye(s)

Signs of an allergic reaction
- Itchy rash (*hives*)
- Swelling of your face, lips or tongue
- Difficulty breathing
- Chest pain or discomfort
- Increase or decrease in your blood pressure (your doctor or nurse will notice this if they are monitoring your blood pressure)

These are most likely during or shortly after the injection.

Signs of a possible liver problem
- Yellowing of your skin or the whites of your eyes
- Unusual darkening of the urine
- Abnormal liver function test

Speak to a doctor or nurse immediately if you get any of the side effects listed above, or if you think you have an infection. **Show your Patient Card** and this package leaflet to any doctor or nurse who treats you, not only to your neurologist.

Other side effects
Very common (may affect more than 1 in 10 people)
- Urinary tract infection
- Sore throat and runny or blocked up nose
- Headache
- Dizziness
- Feeling sick (*nausea*)
- Joint pain
- Tiredness

Common (may affect up to 1 in 10 people)
- Anaemia (decrease in your red blood cells which can make your skin pale and can make you feel breathless or lacking energy)
- Allergy (*hypersensitivity*)
- Shivering
- Itchy rash (*hives*)
- Being sick (*vomiting*)
- Fever
- Difficulty breathing (*dyspnoea*)
- Reddening of the face or body (*flushing*)
- Herpes infections
- Discomfort around the place you have been injected. You could experience pain, bruising, redness, itching or swelling

Uncommon (may affect up to 1 in 100 people)
- Severe allergy (*anaphylactic reaction*)
- Progressive multifocal leukoencephalopathy (PML)
- Inflammatory disorder after discontinuation of the medicinal product
- Facial swelling
- An increase in the number of white blood cells (*eosinophilia*)
- Reduction in blood platelets
- Easy bruising (purpura)

Rare (may affect up to 1 in 1 000 people)
- Herpes infection in the eye
- Severe anaemia (decrease in your red blood cells which can make your skin pale and can make you feel breathless or lacking energy)
- Severe swelling under the skin
- High levels of bilirubin in the blood (*hyperbilirubinaemia)* which may cause symptoms such as yellowing of your eyes or skin, fever and tiredness

Not known (frequency cannot be estimated from the available data**)**
- Unusual infections of brain and eyes
- Damage to your liver

Speak to your doctor as soon as possible if you think you have an infection. You will also find this information in the Patient Card you have been given by your doctor.

Reporting of side effects
If you get any side effects, talk to your doctor. This includes any possible side effects not listed in this leaflet. You can also report side effects directly via the national reporting system listed in Appendix V. By reporting side effects you can help provide more information on the safety of this medicine.

5. How to store Tysabri

Keep this medicine out of the sight and reach of children.

Do not use this medicine after the expiry date stated on the label and carton. The expiry date refers to the last day of that month.

Store in a refrigerator (2 °C - 8 °C).
Do not freeze.
Keep the syringes in the outer carton in order to protect from light.

The pre-filled syringes can be kept at room temperature (up to 30 °C) for a maximum of a combined period of up to 24 hours, including the time to allow warming to room temperature for administration. The syringes can be returned to the refrigerator and used before the expiry date stated on the label and carton. Date and time of removal of the pack from the refrigerator must be recorded on the carton. Discard the syringes if left out of the refrigerator for more than 24 hours. Do not use external heat sources such as hot water to warm the pre-filled syringes.

Do not use this medicine if you notice particles in the liquid and/or the liquid in the syringe is discoloured.

6. Contents of the pack and other information

What Tysabri contains
The active substance is natalizumab.
Each 1 mL pre-filled syringe contains 150 mg natalizumab.

The other ingredients are:
Sodium phosphate, monobasic, monohydrate,
Sodium phosphate, dibasic, heptahydrate,
Sodium chloride (see section 2 ‘Tysabri contains sodium’),
Polysorbate 80 (E 433)
Water for injections

What Tysabri looks like and contents of the pack
Tysabri is a colourless to slightly yellow, slightly opalescent to opalescent (pearly) liquid.
Each carton contains two syringes.
Tysabri is available in packs containing 2 pre-filled syringes or 6 pre-filled syringes.
Not all pack sizes may be marketed.

Marketing Authorisation Holder and Manufacturer
Biogen Netherlands B.V.
Prins Mauritslaan 13
1171 LP Badhoevedorp
The Netherlands

For any further information about this medicine, please contact the local representative of the Marketing Authorisation Holder.

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România
Johnson & Johnson Romania S.R.L.
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Slovenija
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Slovenská republika
Biogen Slovakia s.r.o.
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Suomi/Finland
Biogen Finland Oy
Puh/Tel: +358 207 401 200

Sverige
Biogen Sweden AB
Tel: +46 8 594 113 60

This leaflet was last revised in .

Other sources of information

Detailed information on this medicine is available on the European Medicines Agency web site: https://www.ema.europa.eu.

INSTRUCTIONS FOR USE

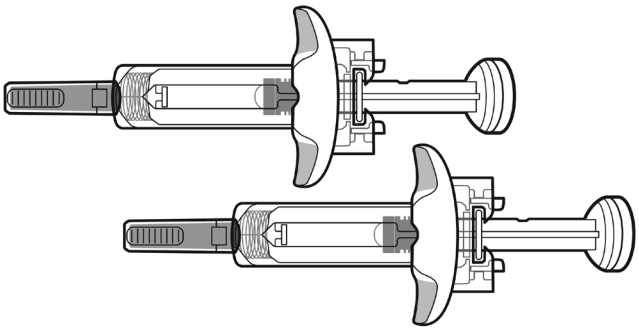
Tysabri 150 mg

solution for injection in pre-filled syringe

natalizumab

subcutaneous use

Full dose = Two pre-filled syringes



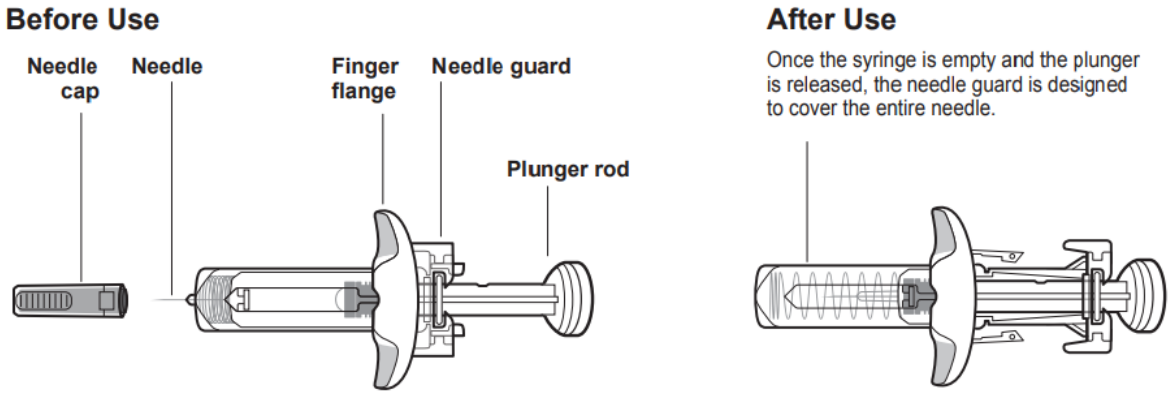
This “Instructions for Use” contains information on how to inject using the Tysabri pre-filled syringe.

Read these Instructions for Use before you start using the Tysabri pre-filled syringe (called “syringe” in these instructions) and each time you get a refill. There may be new information.

This information does not take the place of talking to your healthcare professional about your medical condition or your treatment.

Tysabri Device Parts

Do not remove the finger flange. The finger flange will allow you to hold the syringe more firmly during the injection process.



Important information you need to know before injecting Tysabri

- Tysabri is provided in a pre-filled syringe (called "syringe" in these instructions). Each Tysabri carton contains two syringes. You will need to use both syringes, within 30 minutes of each other, to get your full dose.
- In case of self-administration or administration by a caregiver, your healthcare professional should show you or your caregiver how to prepare and inject the syringes before you use them the first time. If you or your caregiver are administering the injections and have missed a dose or have injected only one syringe, contact your pharmacist or treating physician.
- The syringes are for subcutaneous injection only (inject directly into the fatty layer under the skin).
- Each syringe can only be used one time (single use). They cannot be reused.
- Do not share your syringes with other people, even if their illness is the same as yours. You may give an infection to them or get an infection from them.

Note for healthcare professionals:

Patients should be **observed during the subcutaneous injections and for 1 hour after** for signs and symptoms of injection reactions including hypersensitivity. **After the first six Tysabri doses,** regardless of route of administration, patients should be observed after subcutaneous injection according to clinical judgement.

Storing Tysabri

- Keep the syringe and all medicines out of sight and reach of children.
- Store the syringes in the refrigerator (2 °C to 8 °C).
- **If needed, syringes may be stored at room temperature (up to 30 °C) and up to 24 hours in total. If the syringes have been out of refrigerator for more than 24 hours, do not use them.**
- Keep the syringes in the original carton to protect them from sunlight.
- Do not freeze the syringes or expose to temperatures above 30 °C.
- The syringes can be returned to the refrigerator and used before the expiry date stated on the label and carton.

Preparing to Inject Tysabri

1 Collect the Supplies.

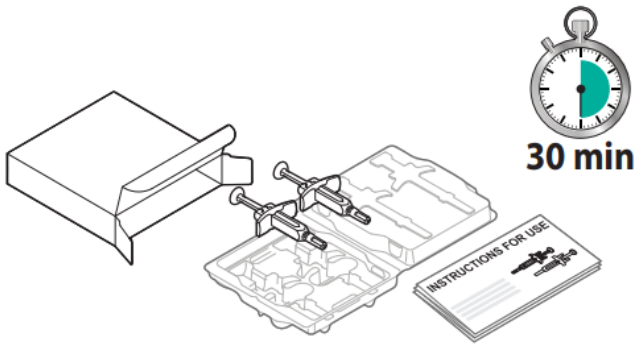
- Collect the supplies and place them on a clean, flat surface in a well-lit area.



2 Remove 2 Syringes from the Refrigerator and Wait 30 Minutes.

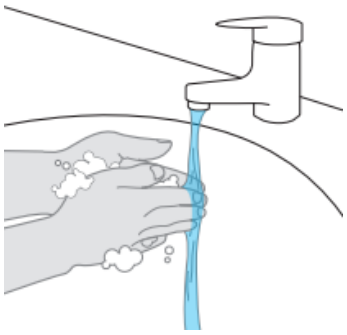
- Take the carton, **containing TWO syringes,** out of the refrigerator and let them come to room temperature (up to 30 °C) for at least 30 minutes.

Do not use external heat sources, such as hot water, to warm the syringes.



3 Wash and Dry Hands.

- Wash your hands well with soap and water. Then dry your hands.



4 Check the Syringes.

a. Check the expiration date on both syringes (callout a).

Do not use the syringe if the expiration date has passed.

b. Check the syringes to see if they are damaged or cracked (callout b).

Do not use the syringe if it is damaged or cracked.

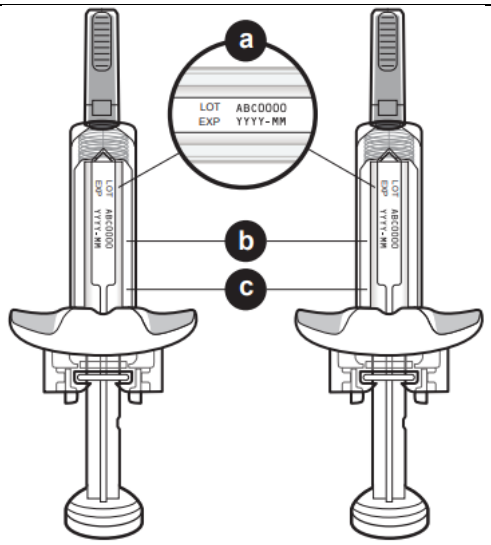
c. Check that the medicine in both syringes is colourless to slightly yellow, slightly opalescent to opalescent (pearly), and free of visible particles (callout c).

Do not use the syringe if the liquid has visible particles.

Do not use the syringe if it has been dropped prior to using it. Notify your healthcare professional if you have any of these issues with the syringes.

You may see bubbles in the medicine. This is normal.

Note: The appearance of the medicine may change after it is removed from the refrigerator. This is normal.



5 Choose the 1st Injection Site.

a. Use one of the following injection sites:

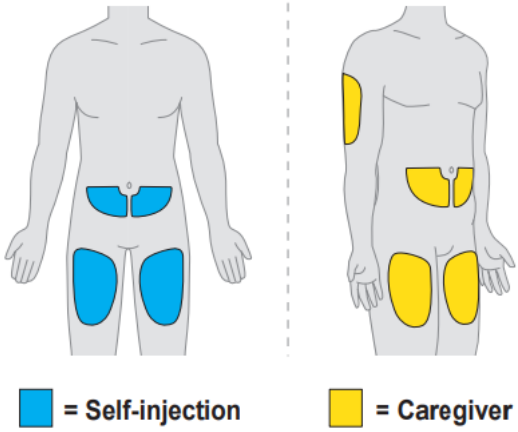
- Abdomen (at least 6 centimetres away from the belly button)

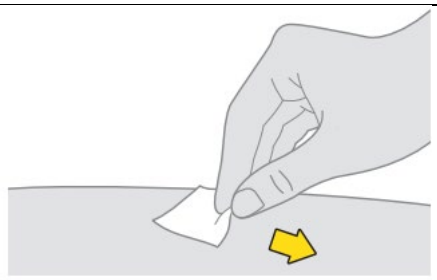
- Front of the thigh

- Upper outer arm (this is for caregiver administration only)

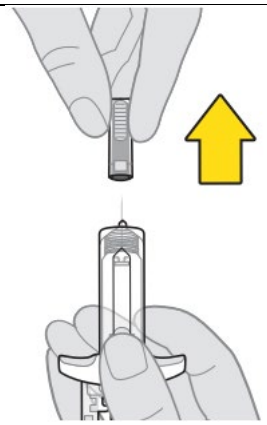
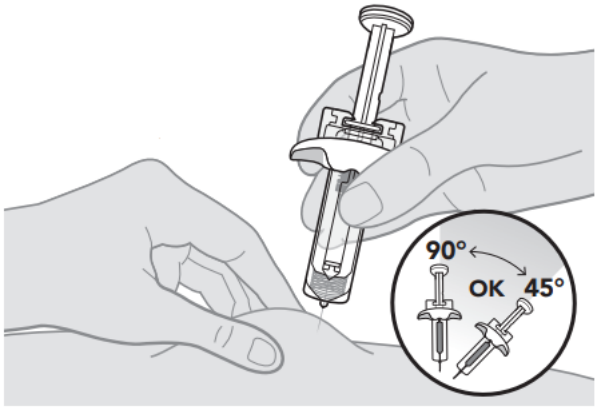
b. **Do not** inject into an area of the body where the skin is tender, red, infected, bruised or has scars.

c. Change (rotate) injection sites between injections.



6 Clean the 1st Injection Site. **a.** Wipe the skin with an alcohol wipe. **b.** Let the injection site air dry before injecting the dose. **Do not** touch, fan or blow on the clean area.	

Injecting Your 1st Syringe

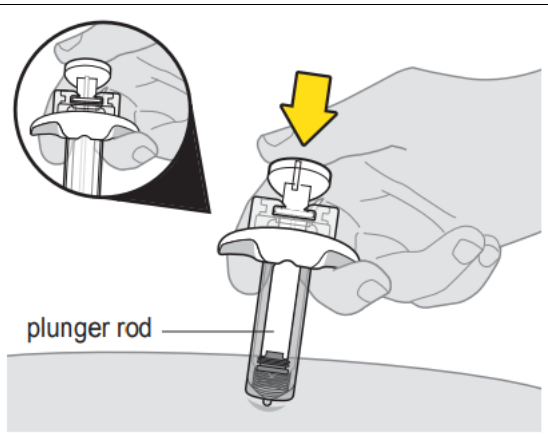
7 Remove the Needle Cap. **a.** Hold the syringe body with one hand with the needle pointing up. **b.** With your other hand, firmly hold the needle cap and pull it straight off the needle. **c.** Dispose of the needle cap right away after removing it. **Note:** you may see a drop of liquid on the needle tip. This is normal. **Do not** touch or re-cap the needle. You could get a needle stick injury. **Do not** pull on the plunger rod.	
8 Insert the Needle into the 1st Site. **a.** Pinch the skin around the cleaned injection site. **b.** With your other hand, hold the syringe like a pencil and use a quick, dart-like motion to insert the needle at an angle of 45 to 90 degrees until the needle is fully under the skin.	

9 Give Your 1st Injection.

a. After the needle is in the skin, release the pinched skin.

b. Slowly push the plunger all the way down as far as it will go to inject all of the medicine.

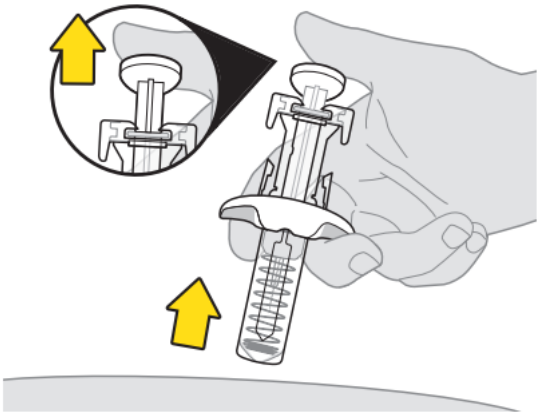
Make sure you push the plunger rod all the way down to get all your medicine and engage the needle guard.



10 Remove the Needle from the Injection Site.

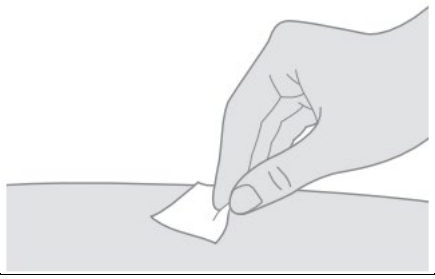
- When the syringe is empty, begin releasing the plunger and remove the syringe from the injection straight out until the entire needle is covered by the needle guard.

If the needle guard does not activate to cover the needle, **do not** recap the syringe. Place it in the sharps disposal container and contact your healthcare professional for assistance.



11 Check and Care for Your Injection Site.

- Apply a gauze pad or adhesive bandage to your injection site, if needed.



Injecting Your 2nd Syringe

12 Choose the 2nd Injection Site.

a. Choose another area to inject. You may use one of the following injection sites:

- Abdomen (at least 6 centimetres away from the belly button)

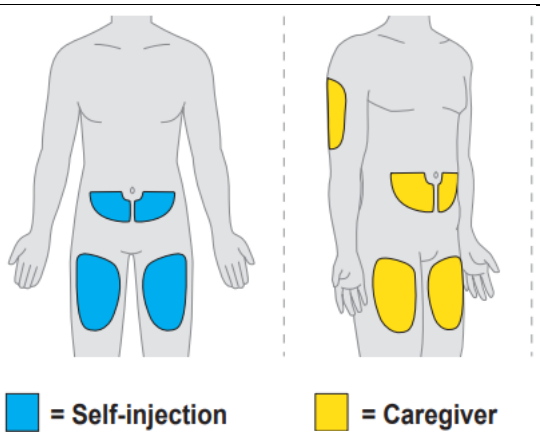
- Front of the thigh

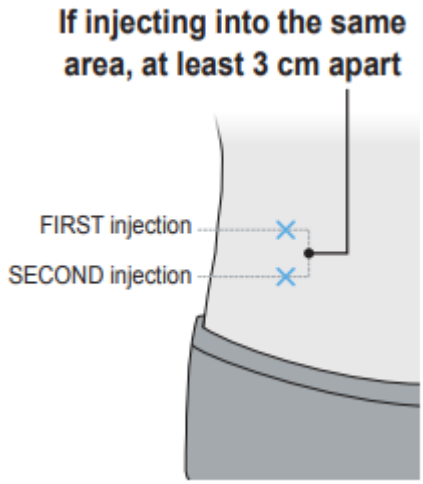
- Upper outer arm (this is for caregiver administration only)

b. **Do not** inject into an area of the body where the skin is tender, red, infected, bruised or has scars.

c. Change (rotate) injection sites between injections.

d. If using the same area of the body, make sure your 2nd injection site is at least 3 centimetres from your first injection site.



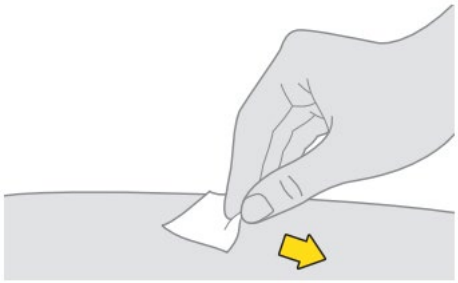


13 Clean the 2nd Injection Site.

a. Wipe the skin with an alcohol wipe.

b. Let the injection site air dry before injecting the dose.

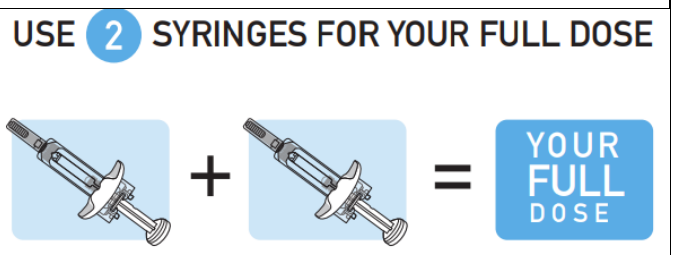
Do not touch, fan or blow on the clean area.



14 Give Your 2nd Injection.

a. **Repeat steps 7-11 to inject the SECOND syringe** to get your full dose.

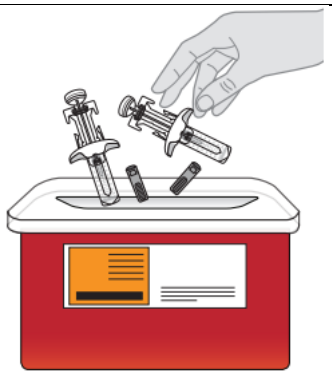
Administer injections one after the other without significant delay. The second injection should be administered no later than 30 minutes after the first injection.



Disposing Tysabri

15 Dispose of Both Syringes.

- Put both used syringes in a sharps disposal container right away after use.



Do not dispose of your used sharps disposal container or any used syringes in your household trash.

If you do not have a sharps disposal container, you can ask your healthcare professional for one or you may use a household container that is:

- made of heavy-duty plastic,
- can be closed with a tight-fitting, puncture resistant lid, without sharps being able to come out,
- upright and stable during use,
- leak-resistant, and
- properly labelled to warn of hazardous waste inside the container.

When your sharps disposal container is almost full, you will need to follow your community guidelines for the right way to dispose of it. There may be state or local laws about how you should dispose of used syringes.

Do not dispose of your used sharps disposal container in your household trash unless your community guidelines permit this.

Do not recycle your used sharps disposal container.